

Virtual Worlds, Real Impact
Mapping EU Strengths and
Global Interdependencies

Abendroth-Dias, K., Soguero Escuer, J., De Prato, G.

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ABSTRACT

Virtual worlds are reshaping the boundaries of human interaction and reality and impacting production process and business models. This report provides a comprehensive analysis of the global and European Union (EU) virtual worlds ecosystem, mapping over 88,000 activities across business, innovation, and research domains led by 68,000 players, including firms, research institutions, and government organisations.

The EU is the global leader in virtual world related research publications. The EU's strengths in foundational research paired with strategies to accelerate innovation-to-market pipelines could foster cross-sector collaboration and address regional fragmentation.

Globally, venture capital funding into virtual worlds reveals stark disparities. The US predictably dominates the the virtual worlds landscape, with roughly EUR 17 billion invested by the end of 2024. China ranks second worldwide with nearly EUR 3 billion invested. While the EU accounts for 17% of the global share of virtual worlds deals, ahead of China (8%) and second to only the US (40%), it comes in fourth behind the UK in terms of the total amount invested. Public funding and cross-border ownership patterns highlight the EU's growing role as both an investor and recipient of foreign capital.

The report identifies significant concentrations of virtual world activities across industrial ecosystems, with Creative Cultural Industries, Tourism, and Retail leading adoption both in the EU and worldwide. However, critical industries like Healthcare, Aerospace & Defence, and Energy-Intensive Industries remain underutilised with opportunities for EU expansion, despite research on their potential impact across these industries. The report also presents the matrix of key enabling technologies like Extended Reality (XR), AI and IoT that comprise global activity in virtual worlds.

By leveraging its research capabilities and fostering strategic alliances, the EU can strengthen its global competitiveness to shape the next generation of virtual worlds.

FOREWORD

A student puts on a headset on and steps into a virtual biology lab to attend a class; a surgeon practices complex surgical simulations in a virtual environment; a person creates and sells virtual experiences in blockchain-based video games. These scenarios are no longer unrealistic: just the opposite, virtual worlds (VW) have evolved to enable real-time co-presence, trainings, simulations, industrial digital twins, augmented learning environments and new forms of cultural and economic participation.

As technologies enabling virtual worlds from extended reality (XR) to artificial intelligence (AI) mature, they transform the way we live, work, and connect. This transformation needs urgent attention from policymakers, innovators, and industry leaders alike.

For the European Union, the stakes are particularly high. We now face a critical strategic question: how can the EU shape this new digital frontier in line with our values, interests and long-term competitiveness? The European Commission strategy on Web 4.0 and Virtual Worlds emphasises the need to steer the next technological transition in a way that reflects EU values and protect people's rights. While these technological transformations bring opportunities – collaboration, creativity, economic growth – they also raise questions about digital sovereignty, control, and values.

We welcome this report providing an evidence-based foundation for understanding the EU's position in the global virtual worlds landscape. This evolving scenario is marked by the EU's **leadership in research** but also a pressing need to **close gaps in innovation** patenting and business activity. With important investment in venture capital, China and US have a consolidated position based on a few global giants. For the EU, **accelerating R&D in enabling technologies** such as extended reality (XR), IoT, blockchain, AI,

5G and 6G and beyond, will be critical to securing a competitive edge. Equally vital is to foster a **vibrant startup ecosystem**, ensuring that EU innovators can scale up without being acquired by foreign giants.

As virtual worlds move from concept to infrastructure, the European Union has an opportunity to reaffirm its leadership in values-driven innovation and to ensure these technologies serve the public interest, economic resilience and democratic governance. The future of virtual worlds is being written today, and research can support the policy choices that will determine how virtual worlds develop in the decades ahead – and whether they reflect Europe's vision of a digital future that is open, fair, sustainable and builds on European sovereignty.



Francesca
Campolongo

Director

Directorate T - Digital
Transformation, AI and
Data
Joint Research Centre
European Commission



Yvo
Volman

Director

Directorate G - Data
Directorate General
for Communication
Networks, Content and
Technology
European Commission

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Authors

- ▶ Abendroth-Dias, Kulani
- ▶ Soguero Escuer, Jorge
- ▶ De Prato, Giuditta

EXECUTIVE SUMMARY

The EU's leadership in virtual world (VW) research contrasts with its lag in business innovation and patent activity, raising risks of dependency on non-EU tech giants and limited influence over global standards, values, and governance in immersive environments. While early-stage sectors like healthcare and mobility show foundational research and patenting potential, this has not yet translated into scalable EU-led ecosystems.

The global landscape is already evolving from virtual worlds (VWs) to augmented physical reality (APR), sometimes called personal/humanist superintelligence, where immersive technologies are increasingly embedded in everyday life. Companies like Meta are investing heavily in AR glasses (e.g., Oculus and upcoming AR headsets) to bridge digital and physical realities, signalling a shift from isolated virtual environments to digitally enhanced physical spaces. This transition presents both opportunities (e.g., industrial applications, smart cities) and risks (e.g., data monopolies, unequal access) for the EU.

The EU's limited ownership of global VW/APR firms and high foreign ownership of EU-based VW companies threaten control over data, IP, and decision-making in critical value chains. European start-ups and SMEs face challenges in securing scale-up funding and avoiding acquisition by dominant non-EU platforms (e.g., Meta, Apple), risking long-term strategic autonomy in VR/XR. The rise of APR technologies (e.g., AR glasses) may marginalize EU efforts if current strategies remain narrowly focused on virtual worlds rather than integrated physical-digital ecosystems.

In presenting these gaps, this report highlights how the EU could tackle these challenges. Prioritising foundations in horizontally integrated innovation ecosystems (e.g. ICT, manufacturing, and retail) could help accelerate

the commercialisation of EU-developed VW/APR technologies. The EU could also look to targeting key sectors (e.g., healthcare, mobility) with sector-specific funding and public-private partnerships to bridge research-business gaps.

Securing strategic assets is integral, and this includes expanding foreign investment screening to protect EU-based VW/APR firms from hostile takeovers and ensuring ownership of critical assets (e.g., IP, data infrastructure). This should go hand in hand with investment incentives and strategic equity participation to promote EU VW business growth.

Internet penetration is a core need to build our virtual worlds. The EU could look to accelerate 5G/6G deployment and semiconductor development (via the European Chips Act) to underpin scalable VW/APR infrastructure. Furthermore, Horizon Europe and Digital Europe Programme funding for SMEs and R&D in key enabling technologies (e.g., advanced displays, optoelectronics) could be scaled in strategic sectors, industrial ecosystems, and patents to support EU virtual world activity.

The EU's Web 4.0 and VW strategy must evolve to address the transition from virtual worlds to augmented physical reality, ensuring European businesses and users thrive in a rights-respecting, interoperable digital future. By addressing ownership gaps, strengthening industrial ecosystems, and leveraging regulatory leadership, the EU can secure a competitive, values-driven position in the global VW/APR landscape.

Main findings

- The EU makes up the third largest share (11%) of global virtual world players, and 9% of research publications, priority patent filings and business operations (collectively called *activities*). China leads with 46% of players and 45% of activities, followed by the US (16% of players, 18% of activities). South Korea comes in fourth with 6% of global VW players and 9% of activities.

- ▶ South Korea (6% players, 9% activities) outperforms the EU in activity intensity (1.5 activities per player vs. 0.82 in the EU), with South Korean players 1.81 times more active on average.
- ▶ Globally, over 68,000 firms, research institutions, and government entities (collectively called players) engage in more than 88,000 research publications, priority patent filings, and business operations related to virtual worlds across the digital ecosystem.
- ▶ The EU leads in virtual world research publications (30% to the US' 16% and China's 15%), driven by Germany, Italy, and Spain, but lags significantly in priority patent filings (just over 1,500 of more than 42,000 global priority filings). This mismatch signals a missed opportunity to translate academic excellence into competitive business assets.
- ▶ Within the EU, Germany (20%), France (14%), and Spain (10%) dominate virtual world activities. Globally, China, the US, UK, and South Korea are the central hubs, with China and the US leading in patents and business activity.
- ▶ The US and China dominate VC funding for virtual worlds, with EUR 16.8 billion and almost EUR 3 billion invested respectively (2009–2024). The EU's smaller share of just over EUR 1 billion focuses on balanced seed-to-late-stage funding.
- ▶ The EU is the second-largest foreign investor worldwide (16%), with Germany, Sweden, and France targeting the UK and US. However, the US (31% global ownership) holds disproportionate control over virtual world firms, raising sovereignty risks. China's domestic investment has reduced foreign ownership from 27% in 2009 to 4% in 2025, signalling shifting global power dynamics.
- ▶ Over 3,000 players and 800 projects have benefitted from EU funding (2009–2025). Within the EU, a majority of public funded virtual world activities are located in Germany, Italy, Spain and France. In comparison to its engagement in virtual worlds activities overall, Greece has also significantly benefitted from EU public funds.
- ▶ XR (85% of activities) is the core of virtual worlds, but AI and 5G related to virtual worlds receive limited attention in EU research and business—despite their critical role in scalability and intelligence. This underinvestment risks hampering competitiveness in next-gen immersive ecosystems.
- ▶ Creative cultural industries (20% globally, 33% in the EU), tourism (14% globally, 27% in the EU), and retail (13% globally, 24% in the EU) are the most active industrial ecosystems. However, high-potential industries like healthcare, aerospace/defence, and energy remain under adopted, despite the EU's relative strength in healthcare research (4% global share).
- ▶ Manufacturing stands out as a growing sector, with virtual world activity exceeding the global average—a strategic opportunity to leverage for industrial modernization.

Related Joint Research Centre work

This report is part of a wider project, *Observing the Evolution of Virtual Worlds (ObsVW)*, which is part of a strategic collaboration between the Joint Research Centre and DG Connect. The context of the project is the European Commission's strategy on Web 4.0 and virtual worlds, which seeks to steer the next technological transition and ensure an open, secure, trustworthy, fair and inclusive digital environment for EU citizens, businesses and public administrations.

The project has three pillars:

- ▶ Pillar 1: Analysing virtual world-related activities and their distribution across industrial ecosystems.
- ▶ Pillar 2: Identifying and systematically analysing applications of virtual worlds.
- ▶ Pillar 3: Examining digital skills gaps in the context of virtual worlds.

Readers may also be interested in:

Abendroth-Dias, K., Soguero Escuer, J., Fernandez-Cruzado, A., De Prato, G., DGTES Handbook: A Snapshot of EU Digital Competitiveness and Dependencies, European Commission, Ispra, 2026, JRC142570 (forthcoming).

De Prato, G., Calza, E., Fabiani, J., Soguero Escuer, J. And Vazquez-Prada Baillet, M., DGTES mapping of the digital ecosystem - Economic insights for EU competitiveness and economic security, European Commission, Ispra, 2024, JRC139065.

Signorelli, S., Fabiani, J. and De Prato, G., ATLAS: An Analytical Tool for Linking and Assessing industrial ecoSystems - Measuring investment, innovation and the EU's role in worldwide industrial ecosystems, Publications Office of the European Union, Luxembourg, 2025, JRC142571.

Hupont Torres, I., Charisi, V., De Prato, G., Pogorzelska, K., Schade, S., Kotsev, A., Sobolewski, M., Duch Brown, N., Calza, E., Dunker, C., Di Girolamo, F., Bellia, M., Hledik, J., Nai Fovino, I. and Vespe, M., Next Generation Virtual Worlds: Societal, Technological, Economic and Policy Challenges for the EU, Publications Office of the European Union, Luxembourg, 2023, doi:10.2760/51579, JRC133757.

Abendroth-Dias, K., Soguero Escuer, J., Fernandez-Cruzado, A., De Prato, G. *From Bytes to business: Strategic EU ownership and dependencies in virtual worlds*, European Commission, Seville, 2025, JRC142876.

01 INTRODUCTION: PROMISES AND PERILS OF VIRTUAL WORLDS

Imagine that it is 2035. A student slips on a lightweight headset and steps into a virtual biology lab. Moments later, she is participating in a training lab exercise that her school could not have afforded in physical form. For these students, virtual worlds have become a bridge to opportunity: granting access to high quality instruction, global collaborations, and immersive environments that spark curiosity and build real-world skills. Virtual worlds have become engines of inclusion across EU countries and regions, boosting research, innovation, productivity, and economic growth, offering low-cost training, telepresence workspaces, cross-border and intraregional knowledge spillovers, and new arenas for cultural and civic participation.

Yet, these promises of virtual worlds come with the perils of economic coercion and dependency, harassment, surveillance, exclusion, and manipulation at scale. The same virtual environment built to support a student's education could include subtle behavioural nudges built into their analytics dashboards to shape student interactions. Student data could be collected without explicit consent. Algorithms could remain opaque. Harassment could occur in shared virtual spaces where enforcement mechanisms lag. The power to empower via virtual worlds could rest with a handful of large companies, undermining global competition and EU digital sovereignty.

The EU's reliance on a handful of dominant foreign tech giants in virtual worlds poses significant risks to strategic autonomy, data sovereignty, and fair competition. The EU risks major supply chain dependencies in IoT and critical hard and soft infrastructure needed to enable virtual worlds on a handful of countries including China, the US, and South Korea. Their dominance risks entrenching technological lock-in, limiting EU businesses' ability to innovate or compete in global markets, and suffer uncertainty in the face of global tensions and volatility. Without proactive measures to diversify ownership and invest in homegrown capabilities, the EU risks ceding influence over the future of

immersive technologies—technologies that are already reshaping work, healthcare, mobility, and even democratic engagement. This underscores the urgency of strengthening industrial ecosystems and strategic equity participation to ensure Europe's virtual worlds reflect its values and serve its citizens.

Virtual worlds thus stand at a critical crossroads. They offer transformative societal benefits, but without evidence-based, thoughtful governance, transparent design, and enforceable protections for rights and safety of users, they may amplify existing inequities and create new vulnerabilities. The good news is that activity in virtual worlds is still growing. The choices made today by the EU could determine whether these immersive digital landscapes become public goods that expand human potential, or fragmented spaces that monopolise power and erode trust.

This report presents the EU virtual worlds landscape at a critical juncture to inform policy decisions on virtual worlds activity at the EU, national, and regional levels. It describes global and EU virtual world business, research, and innovation patenting activity to build and analyse the picture of EU global competitiveness in this domain. It further offers insights on EU ownership of foreign virtual world firms, as well as foreign stakes in EU virtual world firms to illustrate dependencies and opportunities for strategic partnerships.

Virtual worlds encompass Extended Reality (XR), Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR). These immersive technologies further draw together multiple enabling infrastructures and technological components, including artificial intelligence (AI), next-generation connectivity (5G and 6G), cloud and edge computing, and the Internet of Things (IoT). As the global race to build the 'next internet paradigm' accelerates, questions of governance, market structure, and public safety have become central to strategic planning across the European Union (EU), the United States (US), China (CN), and other major economies. This report presents insights on how

these key enabling technologies support virtual worlds, and how virtual world business, research, and innovation patenting activities are distributed across major geographic players including the EU, US, China, and South Korea.

The United States relies on a predominantly market-driven ecosystem in which major private-sector firms steer the direction of XR development. China is a key player in XR as well, along with South Korea, particularly with regard to patenting activity. Chinese technology giants—including Huawei, Tencent, Baidu, and ByteDance—play strategic roles in linking VR development to cloud platforms, AI frameworks, and 5G networks. China's approach blends state steering, national standard-setting, and strict content regulation.

The global virtual worlds market is shaped by a handful of dominant firms. Meta is building out expansive platforms designed to create an all-in-one social, commercial, and immersive ecosystems. Apple's Vision Pro signals a shift toward premium mixed reality systems emphasising spatial computing and enterprise integration. Sony remains a major force in gaming through PlayStation VR2. Microsoft's HoloLens and HTC's Vive devices anchor the enterprise and industrial segments, while Chinese firms such as Pico and DPVR maintain a strong presence in their domestic market.

Core technologies are also shaped by foundational software developers and network enablers. NVIDIA provides essential processing power and AI-driven rendering tools, while Unity and Unreal Engine underpin much of the immersive content creation ecosystem. European firms such as Siemens, Dassault Systèmes, Bosch, and Varjo play important roles in industrial simulation, high-fidelity headsets, digital twins, and engineering applications. Telecom innovators—including Ericsson, Nokia, Huawei, ZTE, and Qualcomm—are critical in advancing 5G and 6G networks optimised for immersive and interactive workloads.

Virtual worlds offer substantial economic benefits, helping industries streamline processes,

reduce costs, and accelerate innovation. Digital twins, for instance, allow manufacturers to model factories and supply chains in real time, improving predictive maintenance and operational planning. In construction and engineering, collaborative virtual design accelerates decision-making and reduces material waste. The entertainment and gaming sectors continue to expand rapidly, driving consumer adoption and advancing the underlying technologies. Environmental sustainability is another area where virtual worlds have made contributions. Remote collaboration tools reduce the need for business travel, and virtual prototyping minimises physical waste in design-heavy industries. As organisations track climate impacts more carefully, virtual approaches to planning, modelling, and communication can help lower carbon footprints. These benefits can be realised in the EU through careful analysis of this growing ecosystem, and targeted policy interventions to maximise gains while mitigating adverse impacts.

A strong European VW ecosystem can support and be supported by strategic autonomy. The global VW market size is expected to grow from EUR 27 billion in 2022 to over EUR 800 billion by 2030 (European Commission, 2023). VW technologies are expected to transform business and employment trends across critical sectors, like healthcare, education, art and design, logistics and manufacturing. This rapid development brings opportunities and risks that the EU needs to address (European Commission, 2023a).

This report supports the Web 4.0 and virtual worlds strategy and the Digital Decade policy programme by: i) exploring how the EU stands in the global virtual worlds domain, identifying the most active countries in business, research and innovation; ii) analysing EU public support in the VW field and the capacity to attract private investment in VW firms compared to other leading countries; including an overview of the EU VW startup landscape and their access to capital; iii) examining strategic dependencies through investments and firm ownership structures; and iv) depicting VW adoption across industrial ecosystems.

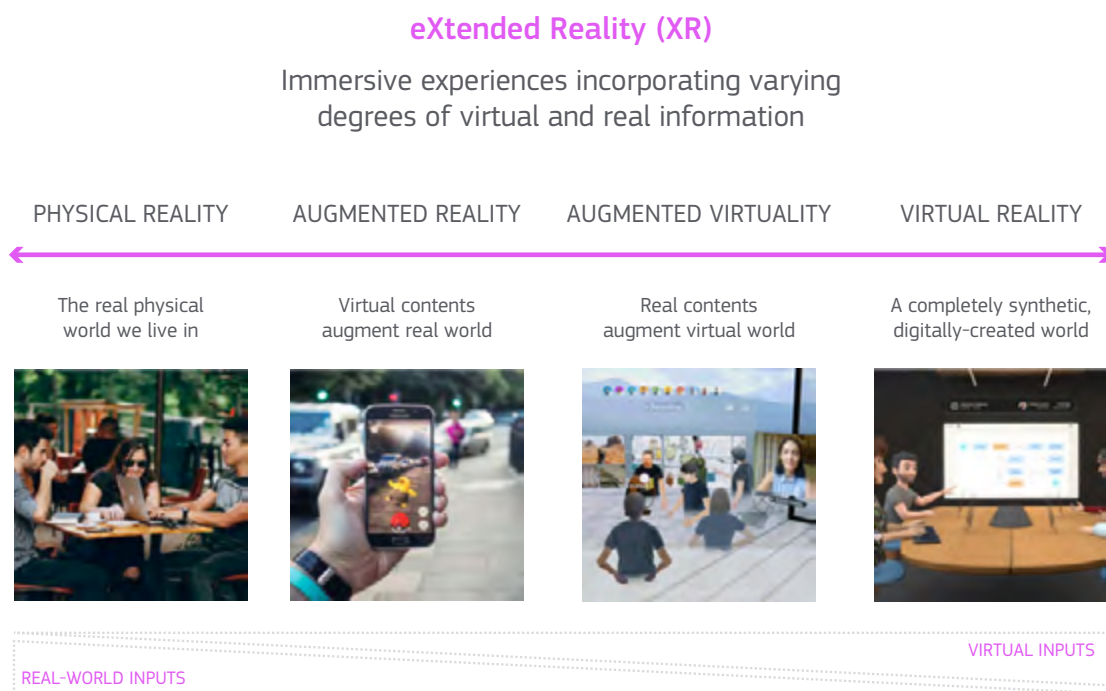
1.1 Definitions and Typologies

Web 4.0 is the fourth generation of the internet that fully integrates physical and virtual worlds and provides better communication and accessibility across devices.

Thus, virtual worlds are a key component of the transition to the Web 4.0. Also known as metaverses, VW refer to computer-simulated immersive environments. Overall, there is a lack of consensus about its definition, but certain

aspects across definitions are recurrently stated, like the virtual-real interplay as well as its societal and human-centric features. (Hupont Torres et al., 2023). The EC defines virtual worlds as ‘persistent, immersive environments based on technologies including 3D and extended reality, which make it possible to blend physical and digital worlds in real-time for purposes such as designing, simulating, collaborating, learning, socialising, transactions, or entertainment’, (European Commission, 2023a).

Figure 1. The reality–virtuality continuum.



Source: Hupont Torres et al. (2023).

To better understand virtual worlds, it is necessary to explain extended reality (XR), which is considered a building block of virtual worlds. The concept of XR leans on the reality–virtuality continuum developed by Milgram and Kishino (Milgram and Kishino, 1994). The boundaries of this continuum are delimited by the physical reality, which corresponds to a fully real environment, and a fully virtual reality. Specific terms refer to technologies within this continuum, depending on the level of blending of real and virtual

environments (Figure 1). XR is the overarching term that covers the set of technologies within the spectrum, that is, all technologies that combine virtual and real environments and human-machine interactions at different degrees (Fast-Berglund et al., 2018). Virtual reality (VR) is an immersive technology embedded in the spectrum of virtual worlds, as it projects an interactive and fully digital environment. VR applications include videogames or applications that generate digital simulations of real-life situations (Vigkos et al., 2022).

Along with VR, augmented reality (AR) is considered a key technology of virtual worlds as allow for innovative immersive experiences. However, AR technologies project digital information into the real world. For example, smartphone camera filters that superimpose computer-generated effects are a form of AR application.

Another important concept is augmented virtuality (AV). AV technologies provide an hybrid but more virtual experience compared to AR (Xi et al., 2023). Mixed reality (MR) usually refers to technologies in the centre of the reality-virtuality continuum, where virtual augments the real world and vice versa. However, MR technologies can also be understood as a highly interactive AR application, where virtual objects realistically blend into, and interact with, real objects and/or with the user (Ecorys, 2021).

1.2 The Digital Techno-Economic Ecosystem Approach

The Digital Techno-Economic EcoSystem (DGTES) methodology combines and harmonises multiple data sources into a network database representation of the digital ecosystem. The figures presented in this analysis includes data released between 2009 and 2025¹. These databases are then used by DGTES to combine and build a wealth of economic indicators and micro-level information, encompassing industrial, geographical, and technologies dimensions.

At the core of the DGTES approach is a system of keywords and queries used to define and generate an ecosystem (and related database) of activities, players, technologies, and interactions (Calza et al., 2022). Keywords and queries reflect the boundaries of the ecosystem to be analysed, so in these cases identify virtual world related technologies.

Activities are identified by several micro-level data sources and categorised into three groups based on their nature and outcomes (De Prato et al., 2019; Righi et al., 2020):

1. **Business activities**, derived from information on a company's core operations, encompassing the production, supply or exchange of goods and services, or investments.
2. **Innovation activities**, represented by priority patent application filings and, with regard to the EU, participation in EU funded R&I projects.
3. **Research activities**, reflecting journal publications.¹

Players are entities that are engaging in the above-mentioned activities across the digital ecosystem, in the sense that they participate in the development, generation, exchange and deployment of knowledge and products (goods or services) within the boundaries of virtual world related technologies. In identifying players, emphasis is placed on organisations rather than individuals i.e. the applicant or developing organisation in the context of patents, affiliation instead of individual authors in conference proceedings and publications. Note that within the scope of this analysis, players and activities were identified for their engagement in activities directly related to VW.

An example of a player is a company registered as producing a virtual reality headset. An example of an activity is a research publication on the impact of virtual worlds on education, or a patent for method and system for eye tracking software.

Players are classified along with the following 3 types:

1. Companies
2. Academic institutions and research centres
3. Governmental authorities and bodies

1. Note that only journal publications published in English are included in this analysis.

Companies with subsidiaries engaged in virtual world activities are captured in this analysis as separate entities i.e. distinct players engaged in the relevant virtual world activity. It may happen that subsidiaries are listed in different countries e.g. a company headquartered in the US which may have a subsidiary in the EU. If the EU-based subsidiary is engaged in a virtual world activity, it is captured as an EU activity.

Business activities are identified via the analysis of micro level data sources, namely Crunchbase, Dealroom, Dow Jones, Orbis, Pitchbook, and Standard & Poor's. Research publications are identified through Scopus. Priority patent filings for innovation activities are identified through Patstat. Activities funded by EU Framework Programmes are identified through Cordis, and cover part of FP7, Horizon 2020 and Horizon Europe.

Thanks to its granularity and flexibility, the DGTES methodology can map a variety of techno-economic segments, fields and topics that have a clear technological connotation, such as virtual worlds. This yields insights into the VW industry, such as the geographical location of VW firms, the technological and industrial composition of their activities, and the structure of foreign ownership or venture capital flows.

To trace the technological perimeter of the VW ecosystem, this methodology considers technology areas associated with enabling or enhancing VW processes. The DGTES approach associates each technology area with a unique set of keywords, which feed into the natural language processing pipeline. In total, this study considers 346 keywords across 7 technology areas: Extended Reality, Artificial Intelligence (AI), 3D Printing, Additive Manufacturing; 5G and Connectivity, Blockchain, Online Platforms, Internet & Cloud, and the Internet of Things (IoT).

Through search queries, these keywords identify documents relevant to our VW definition from

various sources.² By implementing the DGTES approach, the documents refer to three types of activities: business activities (containing information on companies' core business or investments); innovation activities (filing of priority patents or participation in EU-funded projects); and research activities (publications in selected journals, reflecting academic contributions to frontier research). This report covers data and documents published between 2009 and 2025.

The identification of actors from the retrieved documents follows a multi-step process that involves cleaning, disambiguation, classification and validation of the data. As mentioned, actors belonging to the same entity but based in different geographical locations are treated as distinct (therefore DGTES treats subsidiaries as different entities from the parent or funding company). ■

2. The DGTES database is regularly updated, with this report including data for the latest release of the data sources (collated as of August 2025). This has to be taken into account when comparing with other DGTES outputs.

02 THE EU AND THE GLOBAL VIRTUAL WORLD LANDSCAPE

The global virtual worlds ecosystem is comprised of over 88,000 activities spanning business, innovation, and research, carried out by over 68,000 firms, research institutions, and government entities (referred to as players in this report).

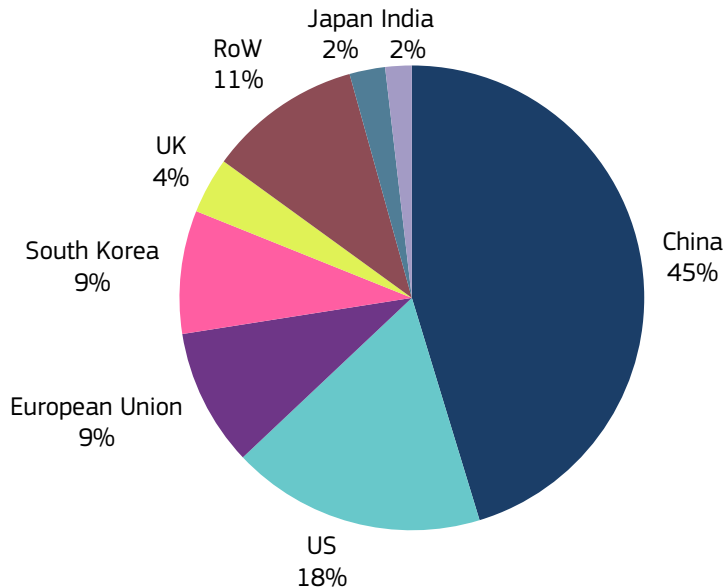
Virtual world activities make up 2% of activities across the global digital ecosystem. Similarly, of the more than 3 million players identified across the digital ecosystem, 2% are engaged in virtual world activity.

A majority of virtual world players are located in China (46%), with the EU coming in third (11%) behind the US (16%). South Korea holds the world's fourth largest share of virtual worlds related players at 6%.

Activities follow a similar trend, with 45% of virtual world activities located in China, 18% in

the US, 9% in the EU, and 9% in South Korea (Figure 2). South Korea generates a higher share of virtual world activities (9%) relative to its player base (6%) compared to the EU, which has a larger player base (11%) but the same activity share (9%). This suggests South Korean players are significantly more active per capita than their EU counterparts. South Korea has 1.5 activities per player (9% activities, 6% players, whereas the EU engages in ~0.82 activities per player (9% activities, 11% players). South Korean players are 1.81 times more active than EU players. This might be linked to multiple factors, including South Korea's exceptionally high fibre adoption (~90.5% of fixed broadband subscriptions are fibre, end-2024) with top-tier fixed broadband penetration among OECD countries—key structural drivers of very high real-world speeds, as well as high engagement in patenting activities (see Section 2.3) (OECD, 2025).

Figure 2. Geographical distribution of virtual worlds activities.



Source: JRC DGTES database.

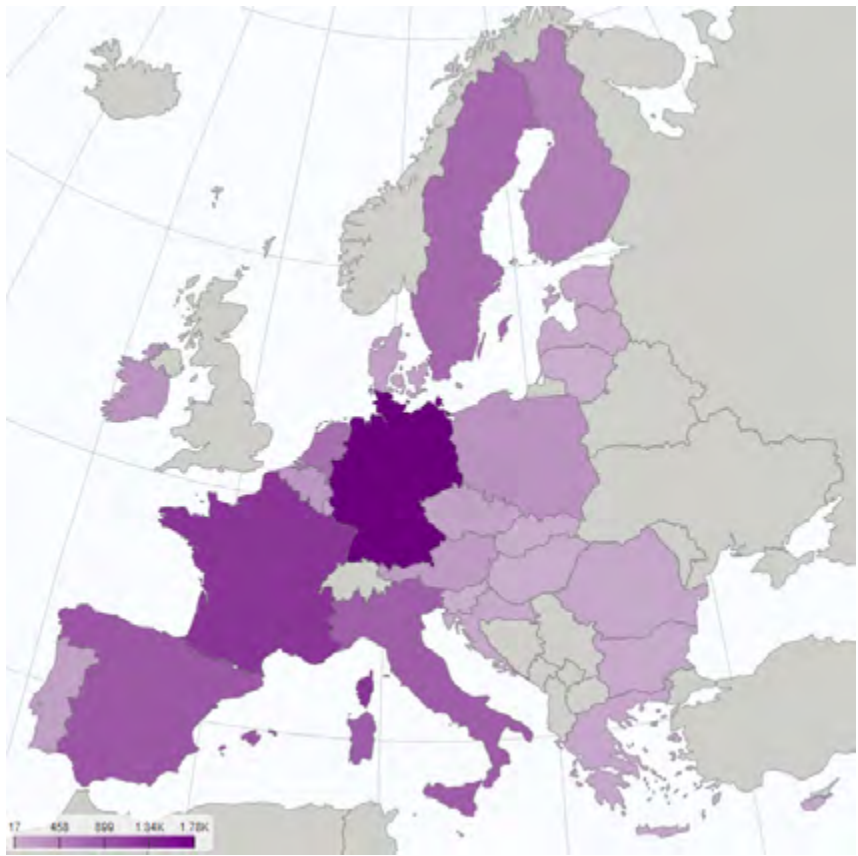
Note: Activities include scientific publications, priority patent filings and business operations.

Within the EU, the largest share of virtual world **players** is located in Germany (17%), followed by France (14%), Spain (12%) and Italy (11%).

Figures 3 and 5 illustrate significant variation in the intensity and concentration of virtual world activity within the EU. By shading countries

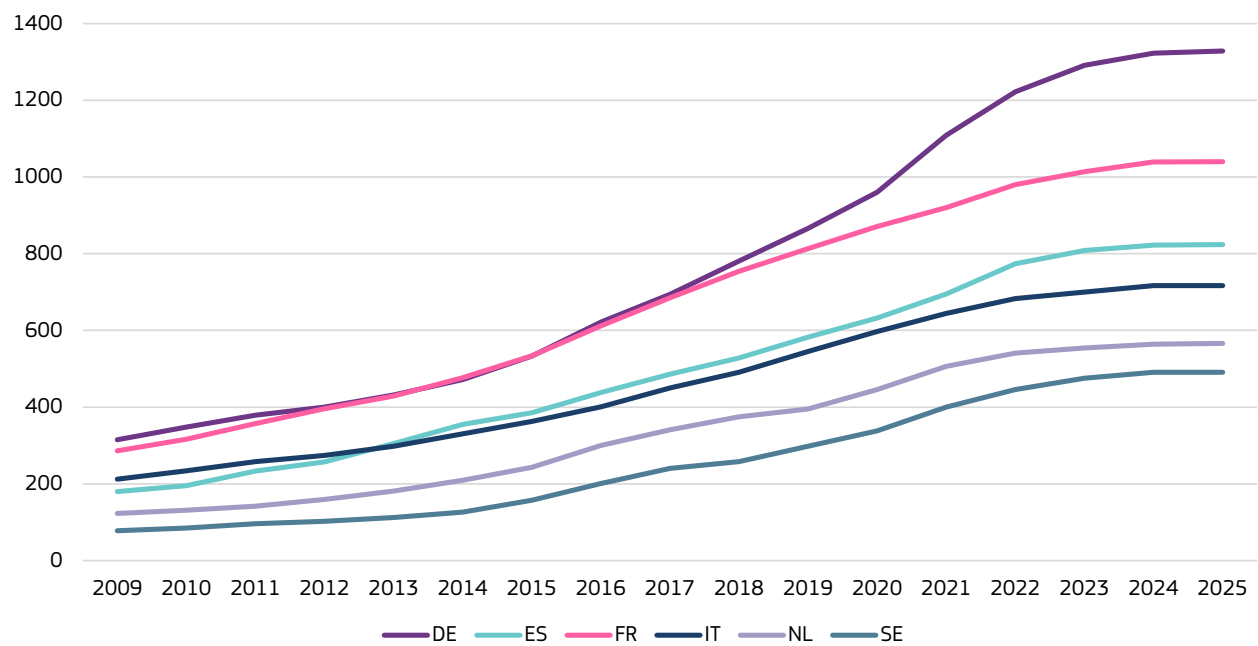
according to their measured level of activity, Figure 3 provides a visual representation of where virtual world capacities are strongest, where growth is emerging. This geographic perspective is essential for informing targeted policy interventions and designing an EU-wide strategy that aligns with national strengths and needs.

Figure 3. Location of virtual world activities in the EU 2009-2025 (cumulative) - country level.



Source: JRC DGTES database.

Figure 4. EU countries with the highest number of virtual world research institutions, businesses, and government organisations 2009-2025.



Source: JRC DGTES database.

Note: EU countries with fewer than 300 players in 2025, or fewer than 3000 players cumulatively over 2009-2025 are not included in this figure.

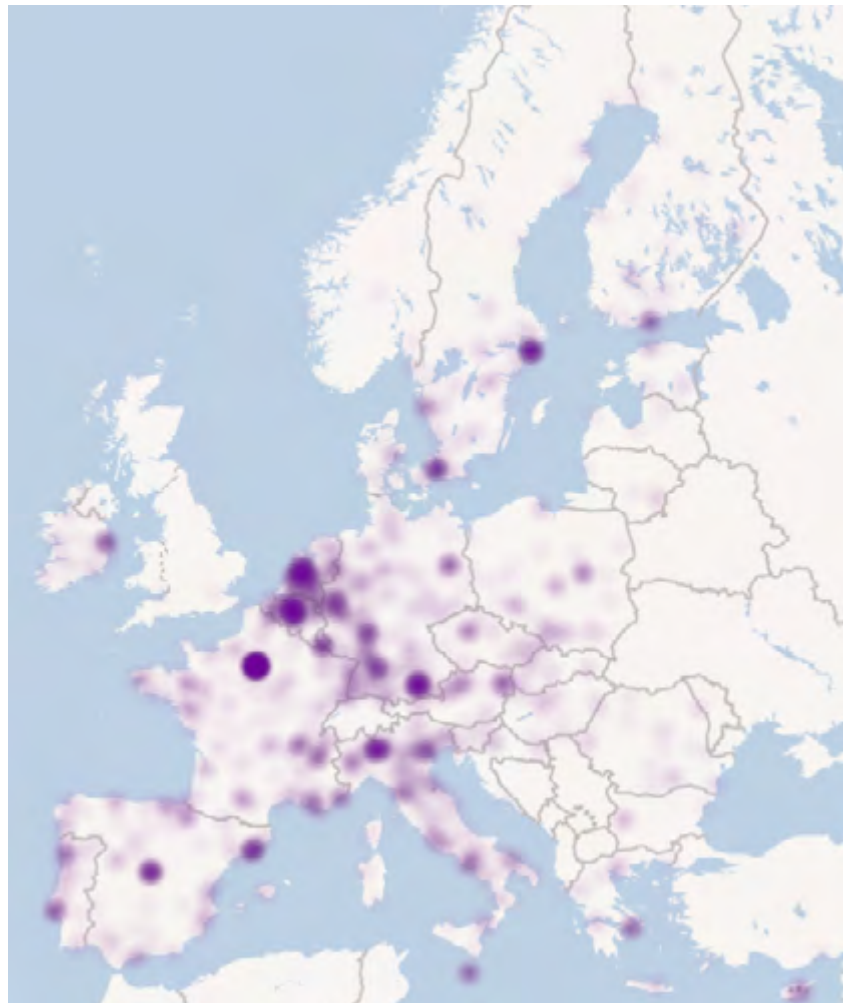
Figures 3 and 5 reveal a somewhat uneven landscape of virtual worlds activity within the EU. A number of countries demonstrate consistently high levels of virtual worlds activity, which may be driven by established research institutions, vibrant technology sectors, access to venture capital, and industrial activity in these locations. Germany shows the highest share of virtual worlds **activity**, followed by France, Spain, Italy, the Netherlands, and Sweden. Within these hubs, virtual world businesses, government organisations, and research institutions have grown gradually, with a marked increase since 2020. In the sections below, we unpack which of these countries are most active in virtual worlds related research, patenting, and business activities.

The map above underscores the need for thoughtful, targeted policy approaches within

the EU to support the emerging distribution of virtual world related activities. High-performing hubs require support to maintain and enhance global competitiveness and deepen cross-border collaboration and support knowledge spillovers. Emerging ecosystems may benefit from attention to scaling tools, access to talent, and regional innovation networks. Countries with limited virtual worlds activity may require structural support to reduce disparities and ensure that the benefits of the virtual world economy are distributed across the EU. This may be possible now due to the still early but growing importance of virtual worlds.

An analysis of virtual world activity at the city level demonstrates that activity distribution exhibits polycentric patterns in some countries while concentrated in others.

Figure 5. Location of virtual world activities in the EU by city 2009-2025.



Source: JRC DGTES database.

Note: Visualisation only includes activities for which city data is available.

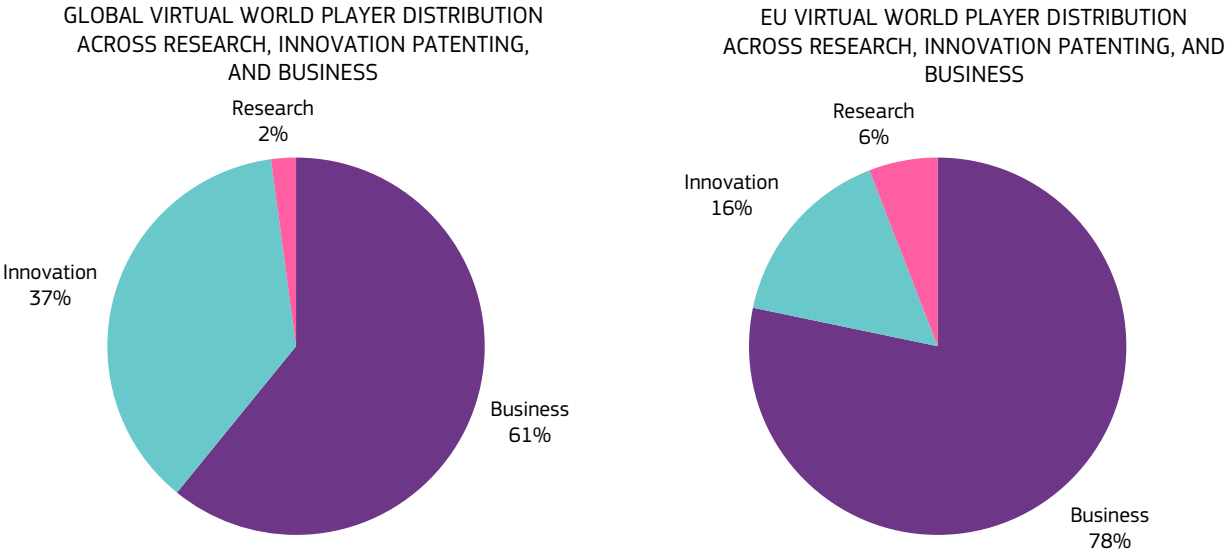
We see that certain EU countries, such as Germany, France, Spain, Italy and the Netherlands have a solid foundation to build out their virtual world activity. Countries such as Germany have polycentric, distributed virtual world-related technological hubs. Others such as France have most of their activity concentrated in the capital.

A majority of EU virtual world players (78%) are engaged in business activities, significantly higher

than the global average of 61%. In contrast, innovation activity is much lower in the EU at 15%, less than half of the global share of 37%.

Meanwhile, the proportion of research players in the EU (6%) exceeds the global average of 2%, highlighting a distinct EU focus on both business and research.

Figure 6. Virtual world player distribution globally versus the EU 2009-2025.

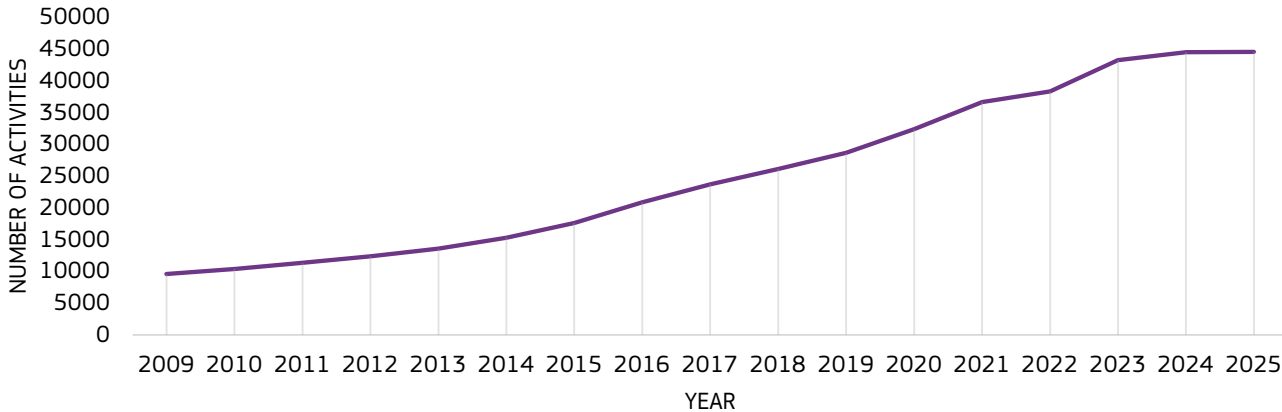


Source: JRC DGTES database.

2.1 Virtual Worlds and Business

Global virtual worlds business activity has expanded between 2009-2025 (see Figure 7). Analysing business activity in this domain is critical to understanding the competitive landscape, investment trends, and economic opportunities emerging in both global and EU-specific markets.

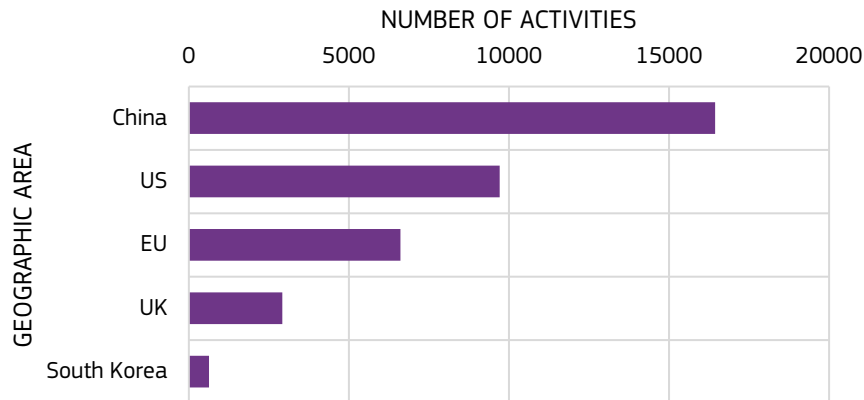
Figure 7. Global virtual world business activity 2009-2025 by year.



Source: JRC DGTES Database.

China has seen rapid expansion in virtual world business activity, followed by the US and EU.

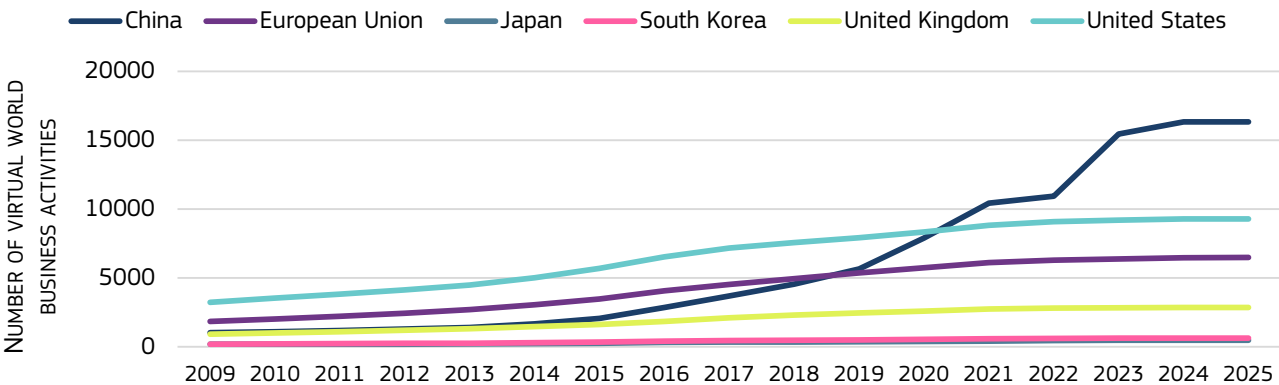
Figure 8. Global virtual world business activity in selected geographic areas 2009-2025.



Source: JRC DGTES database.

Note: Geographic area selection was based on countries most active in virtual world related activity in the business domain.

Figure 9. Growth of virtual world business activity in selected geographic areas 2009-2025.

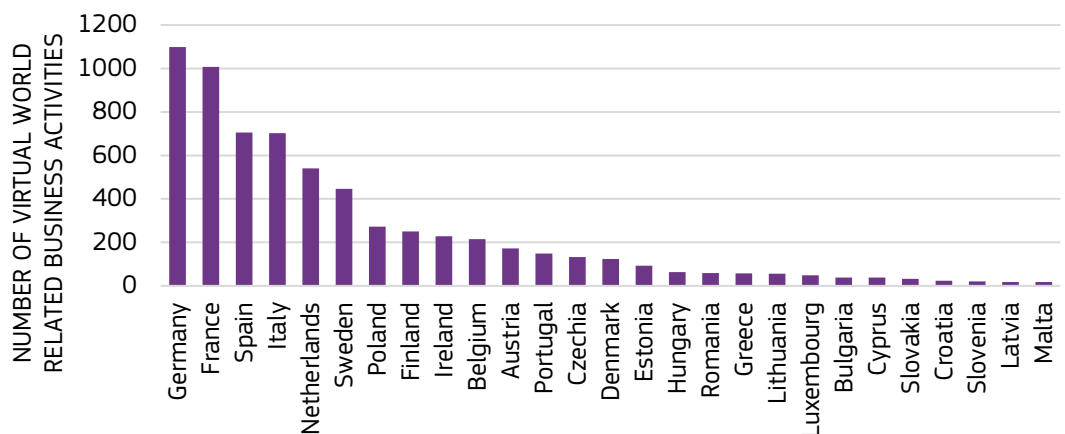


Source: JRC DGTES database.

Note: Geographic area selection was based on countries most active in virtual world related activity in the business domain.

In terms of virtual world related business activities in the EU, Germany leads with 17%, followed by France (15%), Spain (11%), and the Netherlands (8%).

Figure 10. EU virtual world business activity by country 2009-2025.



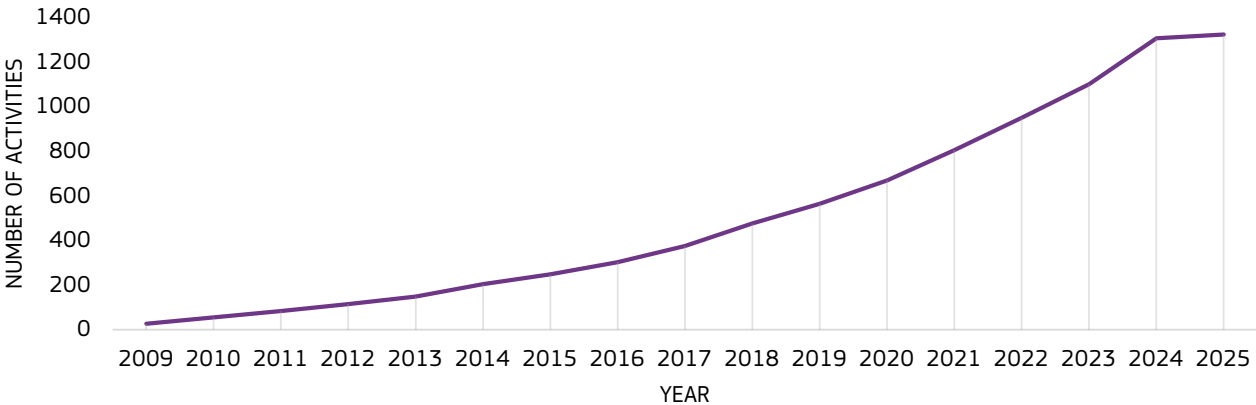
Source: JRC DGTES database.

2.2 Virtual Worlds and Research

Research activity in virtual worlds is a cornerstone of this technology's advancement, shaping foundational capacities of immersive environments. Figure 11 shows how global virtual

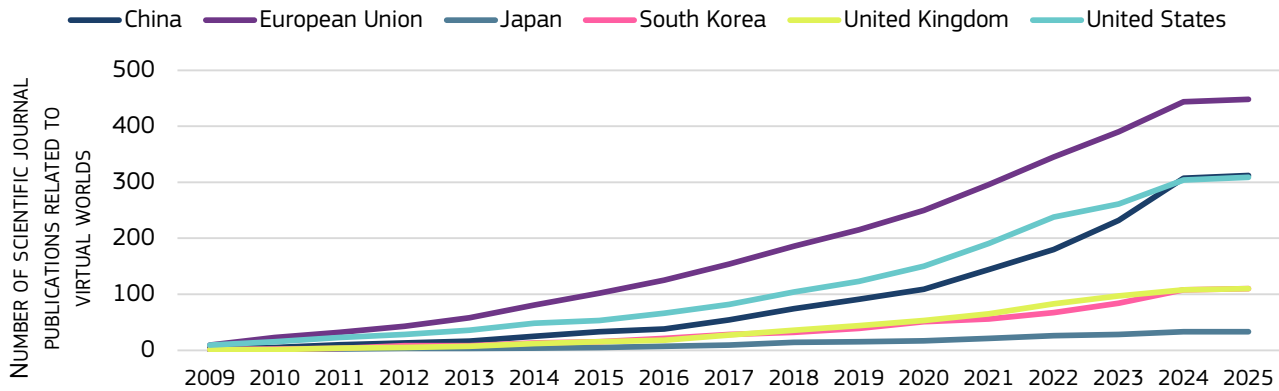
world research publications have expanded between 2009-2025. The EU leads in the global production of virtual world related research publications, indicating a strong foundation to strengthen its global competitiveness in this domain (Figure 12).

Figure 11. Global virtual world research publications 2009-2025 by year.



Source: JRC DGTES database.

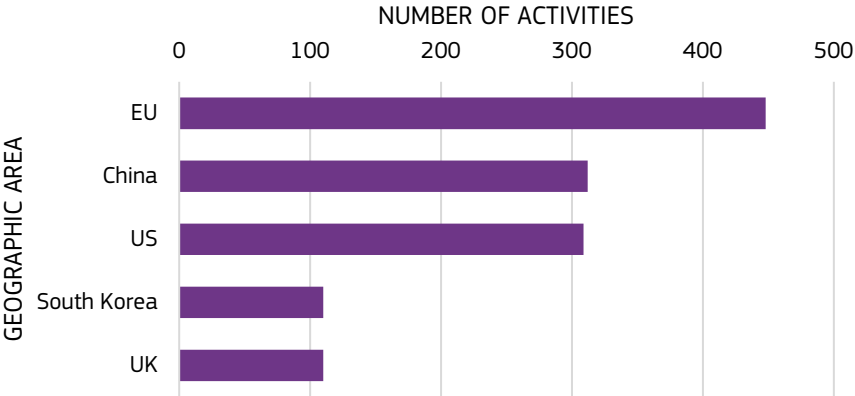
Figure 12. Growth of virtual world research activity in selected geographic areas 2009-2025.



Source: JRC DGTES database.

Note: Geographic area selection was based on countries most active in virtual world related activity in the research domain.

Figure 13. Global virtual world research publications in selected geographic areas 2009-2025.



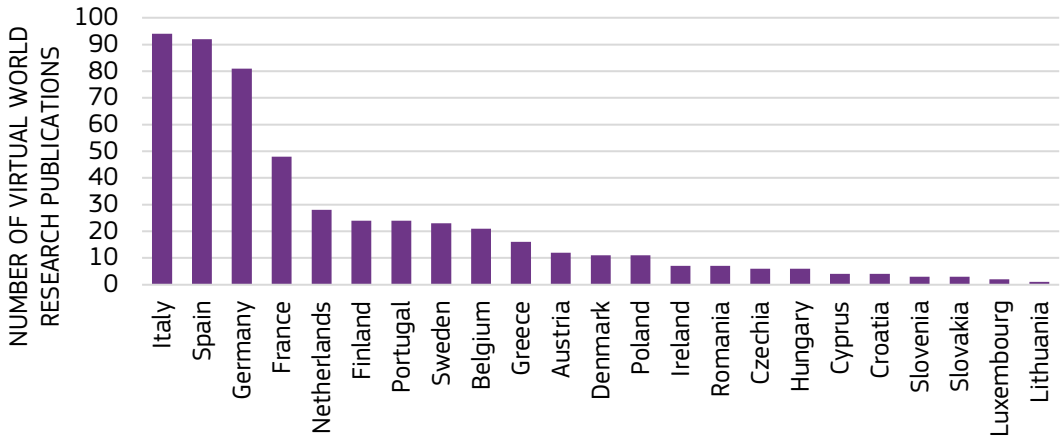
Source: JRC DGTES database.

Note: Geographic area selection was based on countries most active in virtual world related research publications.

Within the EU, Italy leads in the production of research publications related to virtual worlds, followed by Spain, Germany, France, and the Netherlands. EU research is highly distributed and academically-driven. While Italy leads in research publications, the top players account for just 25% of research publications, indicating a broadly distributed, academically-driven research ecosystem rather than a concentrated, firm-led one. Specifically, Politecnico di Milano (IT), Tampere

University (FI), Delft University of Technology (NL), Università Politecnica delle Marche (IT), Universidad de Salamanca (ES), Universitat de València (ES), Technical University of Munich (TUM) (DE), University of Milan-Bicocca (IT), Graz University of Technology (AT), University of Calabria (IT), Universidad Carlos III de Madrid (ES), Universidade de Lisboa (PT), University of Bologna (IT) and INRIA (FR) account for the highest number of research publications related to virtual worlds.

Figure 14. EU virtual world research publications by country 2009-2025.



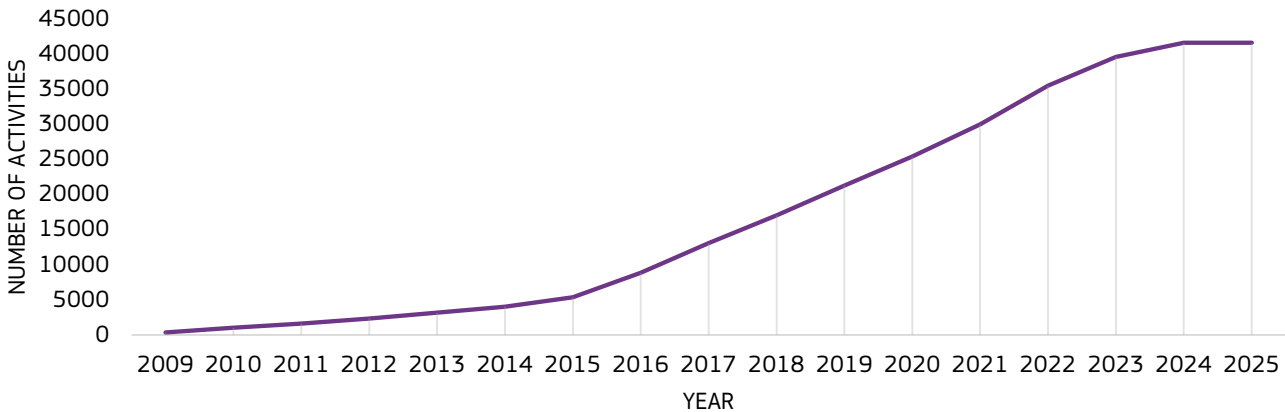
Source: JRC DGTES database.

Note: Only countries with research publications are included above.

2.3 Virtual Worlds and Innovation

Patenting activity in virtual worlds reflects the pace and direction of technological innovation, serving also as a key indicator of intellectual property strategy. Competitiveness in patenting can have global impacts.

Figure 15. Global virtual world priority patent filings 2009-2025 by year.

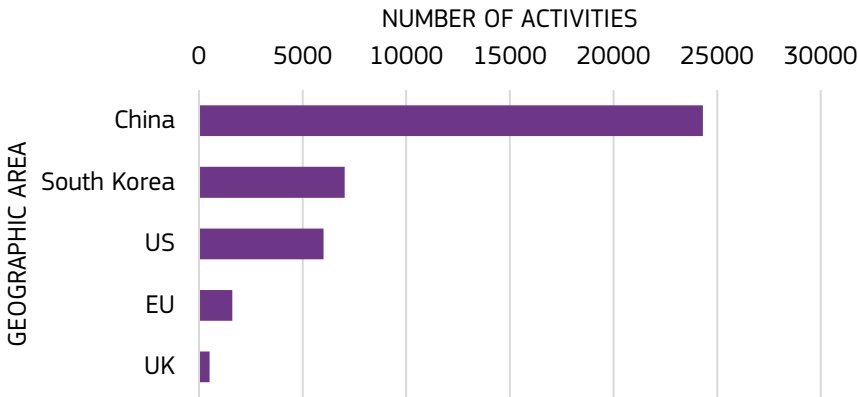


Source: JRC DGTES database.

While global priority patent filings related to virtual worlds have increased, particularly since 2015, the EU comes in fourth behind the US in the global share of these filings. Of nearly 42,000 virtual world-related priority patent filings

between 2009-2025, just over 1500 were filed in the EU. South Korea surpasses the US in their share of priority patent filings, only coming up behind China (Figure 16).

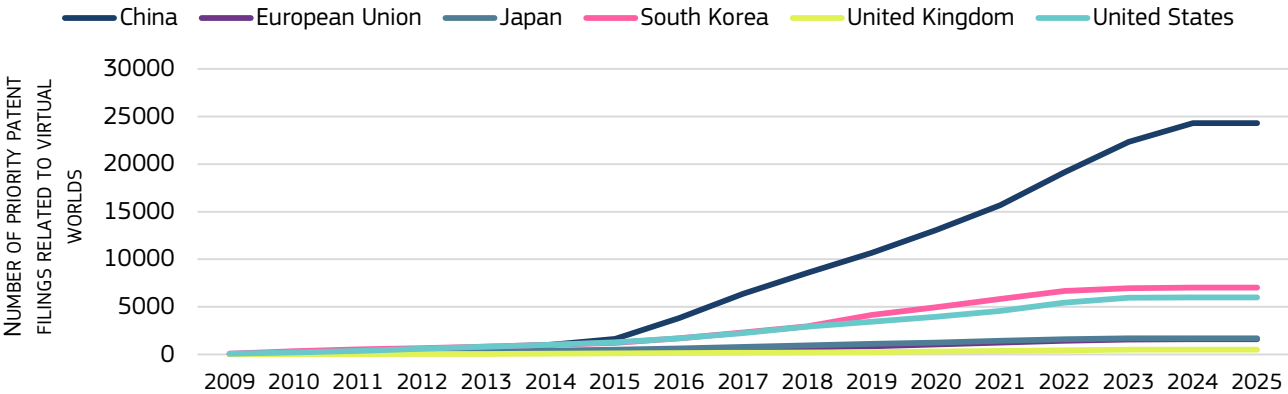
Figure 16. Global virtual world priority patent filings in selected geographic areas 2009-2025.



Source: JRC DGTES database. Author’s own elaboration.

Note: Geographic area selection was based on countries most active in virtual world related priority patent filings.

Figure 17. Growth of virtual world priority patent filings in selected geographic areas 2009-2025.



Source: JRC DGTES database.

Note: Geographic area selection was based on countries most active in virtual world related activity in the patenting domain.

South Korea narrowly surpassing the US might be due to a number of reasons. This analysis counts priority patent applications related to virtual worlds in absolute numbers. The two countries may follow different patenting strategies with South Korea patenting early, broadly, and often, and US players patenting more selectively, relying on software copyright and platform leadership. Virtual worlds patents are also heavily related to hardware: displayers (OLED, micro-LED, semiconductors, GPUs, sensors etc.). South Korea demonstrates leadership across the hardware

stack, which generates patent-dense innovation and leads to many incremental but patentable advances (McFaul, 2023).

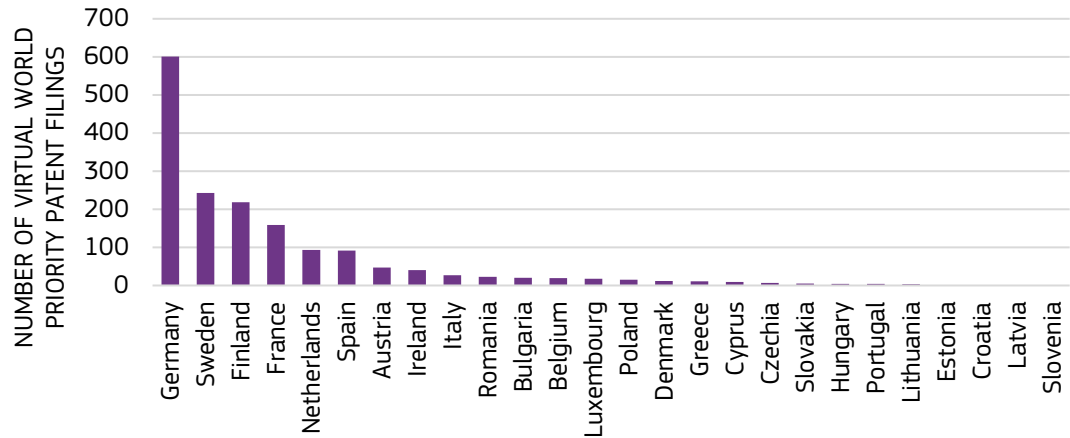
An analysis of the South Korean patenting players shows that patenting activity is highly concentrated: the top 5% of players generate approximately 47% of patents, indicating a relatively small set of players accounting for a large share of patents. LG Electronics, Samsung Electronics, ETRI (Electronics and Telecommunications Research Institute), Korea

Electronics Technology Institute, Bizmodeline Co Ltd, and P&C Solution are the top players generating patents related to virtual worlds in South Korea.

Looking at the EU, the largest share of priority patent filings are made in Germany, followed by Sweden, Finland, and France. An analysis of the players patenting shows that the EU patenting activity is highly concentrated amongst a handful

of players: the top 20 players account for approximately 48% of all entries. Nokia, Ericsson, Porsche AG, Volkswagen AG, and Siemens are among the top players holding patents related to virtual worlds. Among the top players engaged in VW-related patenting, only a small slice are *not* large incumbents. For example, Varjo Technologies Oy is a Finnish manufacturer of virtual reality and augmented reality headsets founded in 2016.

Figure 18. EU virtual world priority patent filings by country 2009-2025.



Source: JRC DGTES database.

Note: Only countries with priority patent filings are included above. Despite the low visibility of line associated with countries in the tail end of the figure, these countries are included in the data and hold a priority patent filing.

In summary, each of these domains— research, innovation and business activity —offers a distinct lens to evaluate the virtual worlds ecosystem. The EU leads the globe in research publications related to virtual worlds. Research activity highlights the drivers of long-term innovation and the alignment of academic priorities with industrial needs. It could build on this strong foundation to attract talent and support industrial research & development.

not in patenting such as Italy, Spain, and the Netherlands. This could enable the EU to build a more resilient, ethical, and globally competitive virtual world ecosystem. ■

Patenting activity provides a measurable snapshot of technological leadership and IP strategy. South Korea stands out as a global leader in this domain, along with China. The EU could look to strengthen its position in patenting activities by building on emerging hubs in Germany, Sweden and Finland, while also building out the capacity of areas that show strength in research and business but

03 STARTUPS, FUNDING AND INVESTMENT

Virtual worlds cannot evolve without sustained investment. Venture capital funding serves as a vital conduit for translating emerging technologies such as Extended Reality, AI, and IoT that comprise virtual worlds into commercial applications. This is especially the case in high-growth, capital intensive fields such as virtual worlds. Public funding plays a pivotal role in bridging the gap between foundational research and their commercialisation. This is particularly the case for the EU, where EU public funding could help support innovation in immersive digital ecosystems and remain aligned with strategic priorities such as digital sovereignty, sustainability, and inclusive growth. The chapter explores the role of EU public funding and venture capital in fostering virtual worlds, analysing investment trends and regional dynamics. This chapter also illustrates the distribution of different types of funding (seed, early- and late-stage, as well as private equity) in the EU, US, China, UK, and South Korea, key players in virtual worlds, to look at support for start-up and scale-up ecosystems within these geographic regions³. Attracting early-stage investment could help firms accelerate product development, gain market entry and expand user bases. Late-stage funding helps enhance operational efficiency and compete in a rapidly evolving global landscape.

The EU has recognised the need to boost investments in all cutting-edge technologies that shape virtual worlds, from XR to blockchain (European Commission, 2023a). The EU Startup and Scaleup Strategy (European Commission, 2025) includes measures aimed at improving access to finance by building a larger, integrated EU VC market and launching a Scaleup Europe Fund to support late-stage growth rounds which could bolster virtual world startup activity in Europe. The EU has looked to accelerating uptake of VW business models

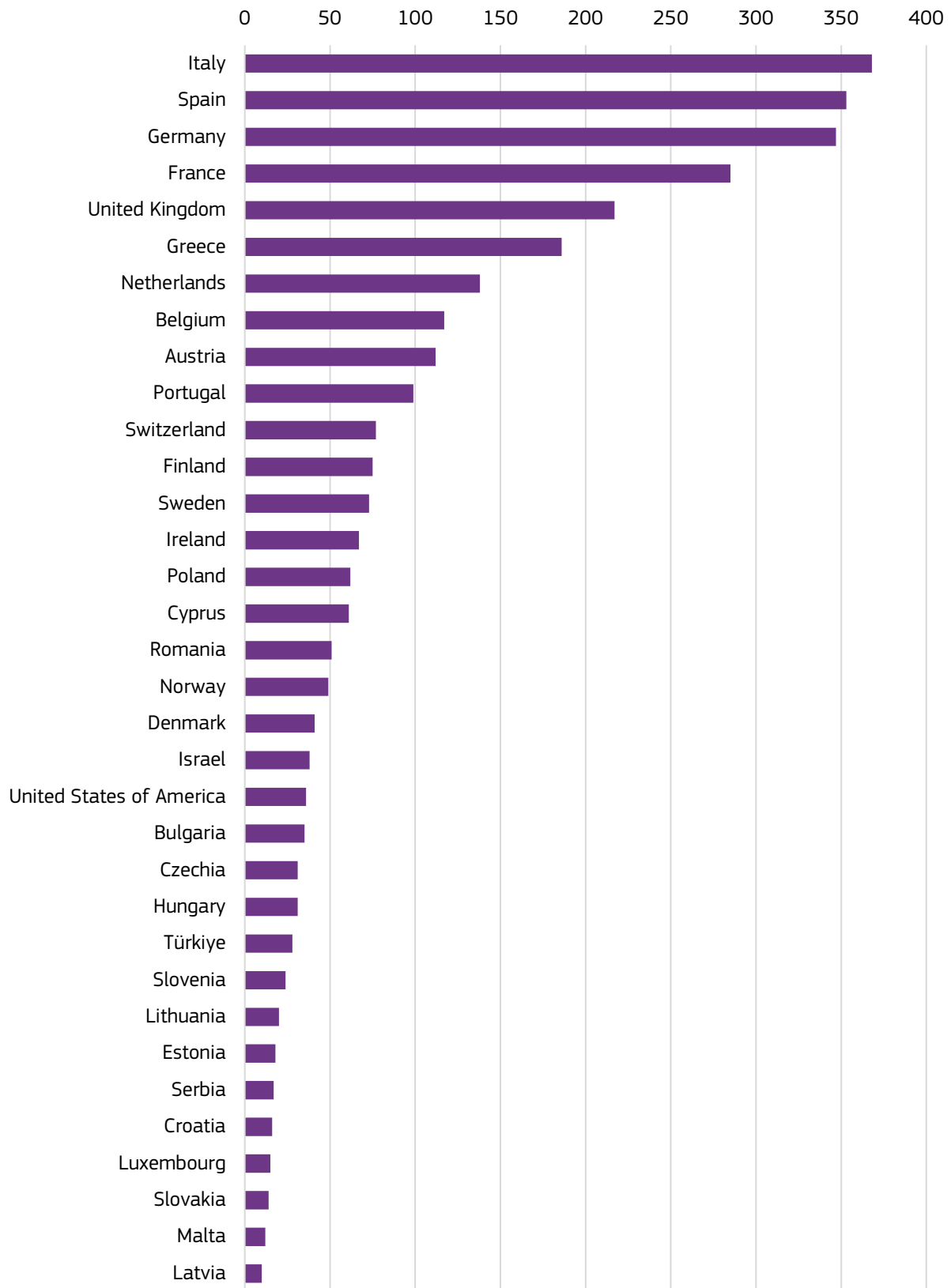
and solutions, connecting developers and industry users, and supporting and scale-up innovative business models, which could all be supported via the Startup and Scaleup Strategy. This section provides insights on where these investments may already have a foundation to perform well and be most critical. Understanding VC and public sector funding trends and their implications is essential to craft strategies that bridge research-business gaps, scale promising innovations, understand regional disparities, and ensure equitable access to the opportunities presented by virtual worlds.

3.1 EU Public Funding of Virtual World Projects

Between 2009-2025, over 3000 business, research and innovation players and over 800 projects have benefitted from EU funded support on virtual worlds. Italy (12%) hosts the highest share of **beneficiaries** of EU-funded projects. Together with Spain (11%), Germany (11%), France (9%) and Greece (6%), they host nearly 50% of the project participants (Figure 19). The share of participants from third countries accounts for 15%. With more than 170 players, the United Kingdom, a former Member State and a Horizon Europe associate, hosts the largest share of beneficiaries not located within the EU, with 6% of virtual world related players receiving EU funds located in the UK, followed by Norway (2%).

3. Some charts include additional countries where they add relevant information. See for example Figure 19. Geographical distribution of VW startups as a share of the global and EU, 2009-2025.

Figure 19. Geographical distribution of beneficiaries of EU-funded projects, 2009-2025.



Source: JRC DGTES database.

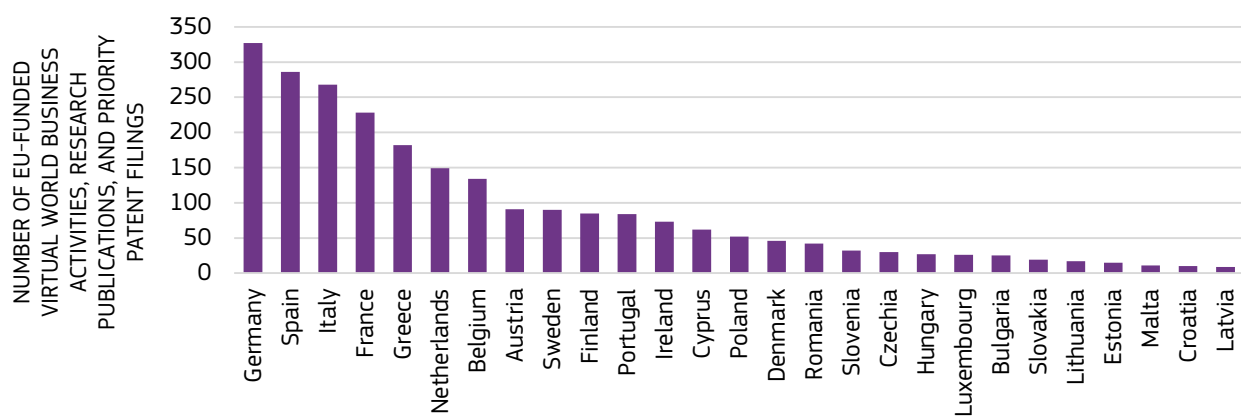
Note: Beneficiaries can refer to businesses, research centres or universities participating in VW-related projects. This analysis includes beneficiaries located in non-EU countries, specifically the UK, Norway, Israel, Türkiye, USA, and Serbia. Non-EU countries with less than 10 beneficiaries have been excluded from this analysis.

The **geographical distribution of projects – each of which usually includes multiple beneficiaries** – provides a slightly different picture. Figure 20 illustrates that a majority of EU-funded projects related to virtual worlds are located in Germany. Spain ranks second while Italy is third. Therefore, while Germany hosts the largest number of EU-funded projects related to virtual worlds, the highest number of beneficiaries

of EU funds related to virtual worlds are Italian. In other words, there is a higher concentration of Italian players working on EU-funded projects.

It is also noteworthy that, relative to the number of virtual worlds players and activities in the country, Greece hosts a significant number of EU-funded virtual worlds players and projects. It comes up fifth just behind France on both accounts.

Figure 20. EU-funded virtual world activities by country 2009-2025.



Source: JRC DGTEs database.

A vast majority of EU-funded virtual worlds projects relate to Extended Reality (84%), followed by AI (7%), Internet of Things (5%), Online Platforms, Internet & Cloud (2%), 5G & Connectivity (2%) and Blockchain (1%). IoT receives a smaller percentage of EU funds relative to its presence as an enabling technology for virtual worlds (see Section 5.2).

3.2 Venture Capital Funding: Global and EU Trends

Venture capital (VC) is a critical driver of innovation, scalability, and market transformation. VC flows are key for competitiveness as they fuel innovation. VC financing provides essential funding and strategic support to startups and small businesses, scaling up growth. They are pivotal for businesses to gain a competitive edge in the digital economy by enabling investment in R&D and new VW technologies. Understanding VC flows also allows policymakers to assess a region's innovative capacity and identify gaps to better target public support strategies.

Figure 21 illustrates the trajectory, in euros, of VC investments in virtual world firms in the EU, US, China, UK, and South Korea. The US predictably dominates the landscape with EUR 16.8 billion invested, driven by peaks in 2016 and 2022. Early investments in virtual worlds between 2013–2016 correspond with the period that Facebook Inc. (now Meta Platforms Inc.) acquired Oculus for approximately 2 billion USD (Facebook, Inc, 2014), and Unity Technologies expanded strategic partnerships with Oculus (Unity Technologies, 2014). Unity began making significant investments in the emerging VR market, including Oculus Rift, HTC Vive, and Samsung Gear VR in 2014–2015 (Allidina, 2025). Niantic released Pokémon Go, in collaboration with the Pokémon Company, in 2016 (The Pokémon Company & Niantic, Inc., 2016). In summary, 2014–2016 saw several US-based companies making strategic investments in the virtual world-related firms.

Investment in virtual worlds technologies peaked again at EUR 3 billion peak in the US in 2022. Epic Games raised US\$1 billion in 2021 and

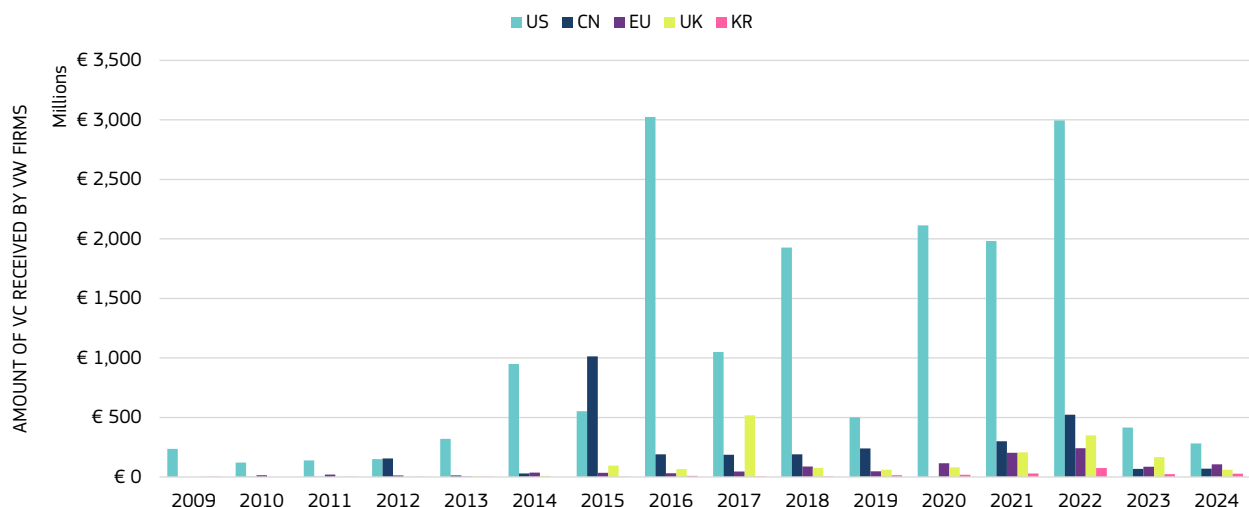
an additional US\$2 billion in 2022 in growth-equity rounds that signalled investor confidence in large-scale, persistent virtual environments (S&P Global Market Intelligence, 2024). The Roblox IPO in 2021, alongside soaring valuations for gaming and creator-platform companies, reinforced the perception that immersive, user-generated virtual worlds were becoming mainstream (McKinsey & Company, 2022; Roblox Corporation, 2021). At the same time, corporate spending surged: Meta dramatically expanded its Reality Labs investments (Dean, 2022), and Microsoft's announcement of the Activision Blizzard acquisition further legitimised virtual world ambitions for both investors and industry analysts (Microsoft Corporation, 2022; Reuters, 2022). This period also coincided with a strong Web3/NFT boom, during which many blockchain gaming and crypto-based virtual-world projects attracted venture funding by branding themselves as 'metaverse' players (CryptoSlate, 2022; Decrypt, 2021; Xiang et al., 2016). Together, these forces created a self-reinforcing cycle of enthusiasm, capital, and media attention that defined the height of the virtual world investment in the US before conditions tightened in 2023.

China ranks second worldwide with nearly EUR 3 billion invested in virtual worlds over the period 2009-2024. Chinese VC investment in virtual worlds

has been more gradual between 2009-2024, with a significant peak in 2015 and increased funding in 2021 and 2022, possibly following market narratives of increased investor confidence in virtual worlds following investments in Epic Games, Roblox, Meta's Reality Labs, and Activision Blizzard by Microsoft, and increased attention to NFTs. While publications discussing Chinese investment in virtual worlds remains rare, the peak in 2015 could be due to the investments made by Meta's Facebook and Unity Technologies into Oculus, HTC Vive, and Samsung Gear VR. Indeed, a majority of Chinese investment related to virtual worlds seems to have been directed towards hardware (i.e. building headsets and related equipment) (Forbes, 2016; TechNode, 2016), that may align with the needs and investments of US-based firms. This period also saw a rapid growth of virtual reality arcades and experience centres across China.

The UK has secured around EUR 1.7 billion in investment in virtual worlds overall between 2009-2024. The EU follows the UK. European firms received the largest amounts of funding in 2020 and 2021, possibly supported by growing US investments into VW during this time (Atomico, 2021), record levels of tech capital invested in the EU more generally in 2021 (Calvino, Criscuolo & Ughi, 2024), and pandemic-driven demand for virtual presence (Ball, Huang & Francis, 2021).

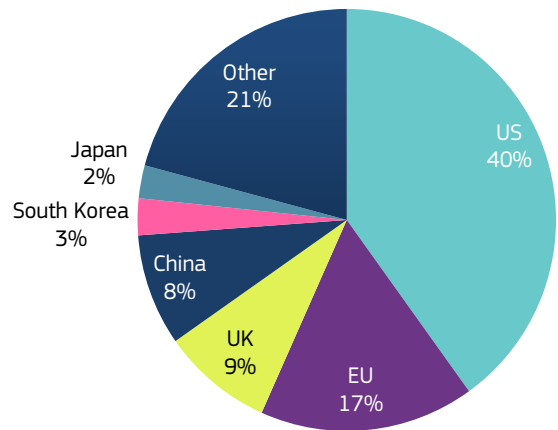
Figure 21. Amount of VC received by VW firms in selected geographic areas 2009-2024 by year.



Source: JRC DGTES database.

Figure 22 illustrates that the EU accounts for 17% of the global share of virtual world deals. It is ahead of China (8%) and second to only the US (40%).⁴ Despite this significant share of global virtual world deals that are carried out in the EU, it still lags in the total amount of VC attracted. The US leads in both the share of VC deals and VC received from 2009-2024, showing strength both in attracting investors and signing large deals.

Figure 22. Geographical distribution of VW-related deals, 2009-2024.

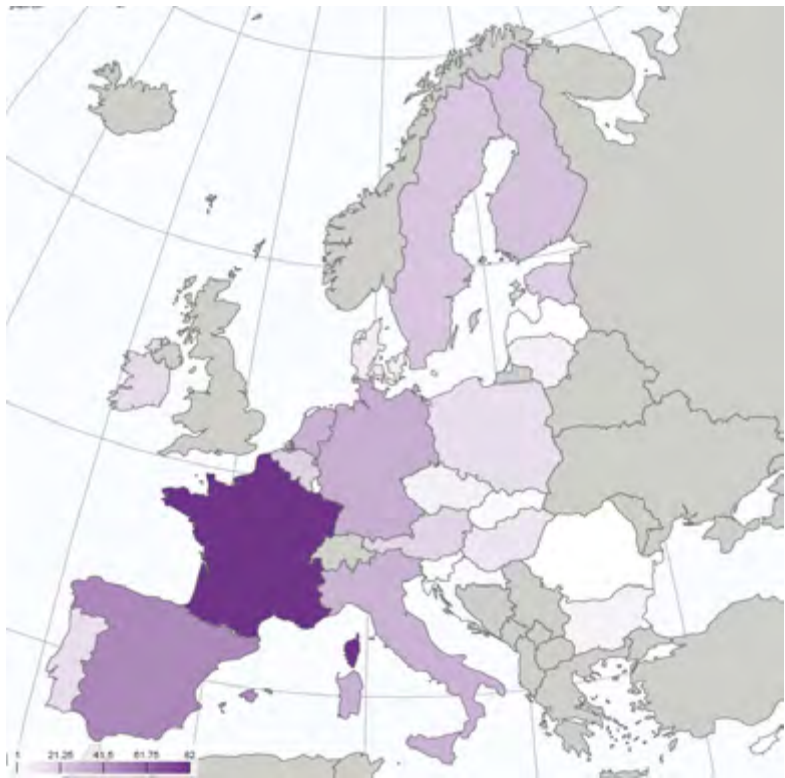


Source: JRC DGTES Database.

Note: This figure includes venture capital (seed, early-stage and late-stage) and private equity deals on VW-related firms.

Within the EU, the majority of deals related to virtual worlds in the EU are attracted by France (19%), followed by Spain (11%), Germany (8%), Italy and the Netherlands at 7% apiece (Figure 23).

Figure 23. Destinations by country of VC deals in the EU 2009-2024.



Source: JRC DGTES database.

4. Analysis of VC data from 2009-2024 across all digital areas in DGTES shows that the EU attracted the fourth highest amount of VC and the second highest amount of VC deals globally. An updated version of this data including 2025 is forthcoming in the DGTES Handbook.

3.3 Virtual World Startups: Global and EU Trends

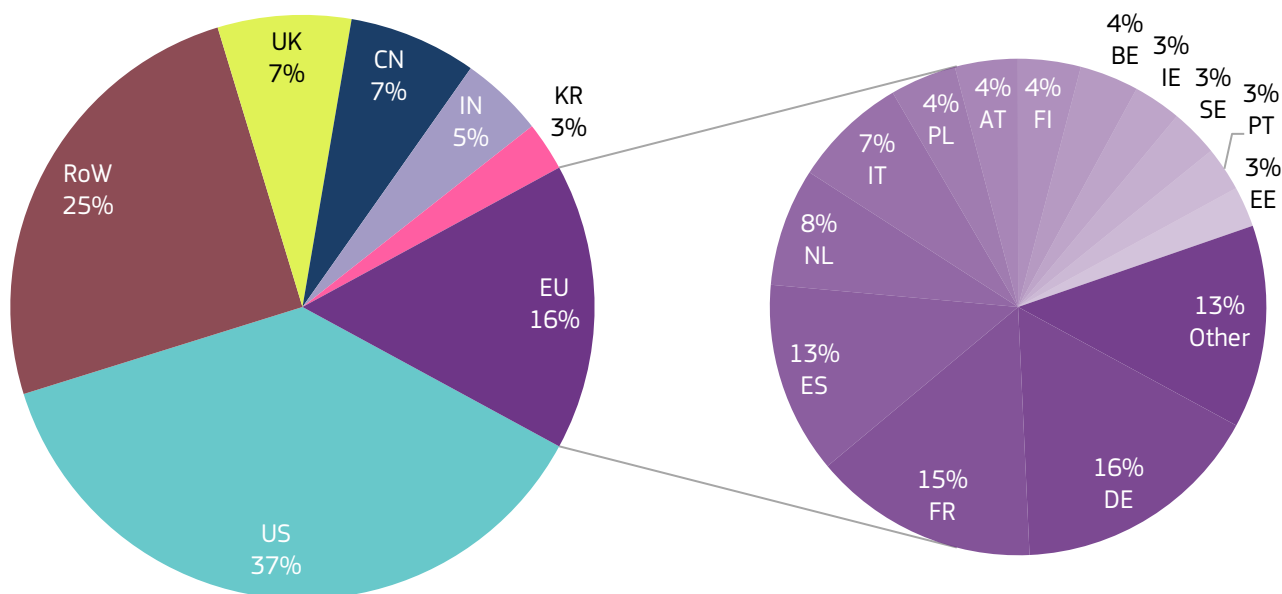
In order to have a better understanding of the VW business environment we dig deeper into the startups' current state. Startups are often seen as high-risk, high-reward businesses and are often associated with innovation and growth. They are also a bellwether of market entry viability, access to financing and business appeal for a given industry.

By 2025, the number of DGTES startups, defined as VW firms that received for the first time an early-stage round (including angel, pre-seed, seed and series A rounds), amounts to over 2500, worldwide. This sum is equal to 6% total VW firms

recorded in the DGTES database. Figure 24 shows the geographical distribution of VW startups. The US (37%) dominates with nearly a thousand virtual worlds young firms. The EU (16%) comes up to the second place with over 400 startups, followed by the UK and China as both account for 7% of the global VW startup landscape.

The distribution of startups across countries is more homogeneous compared to total sum of players –which includes all firms, research centres and government institutions. For example, China does not show a dominant position, while countries like India, Canada, and other Asian, European and Gulf –aggregated under 'Rest of the World'– countries exhibit a dynamic startup environment.

Figure 24. Geographical distribution of VW startups as a share of the global and EU, 2009-2025.



Source: JRC DGTES database.

Note: Number of virtual world firms that received at least an early-stage round (including seed rounds), between 2009 and 2025. RoW refers to rest of the world.

At EU level, the distribution of VW startups is similar to that of the aggregated number of firms across the digital ecosystem. As with digital firms overall, Germany dominates with nearly 16% of the EU share, and equal to 3% of the global share. France (15%) and Spain (13%) come close, and they rank second and third as in the total number of EU players. This is a slight departure when compared to firms across the digital ecosystem, where Sweden comes in third

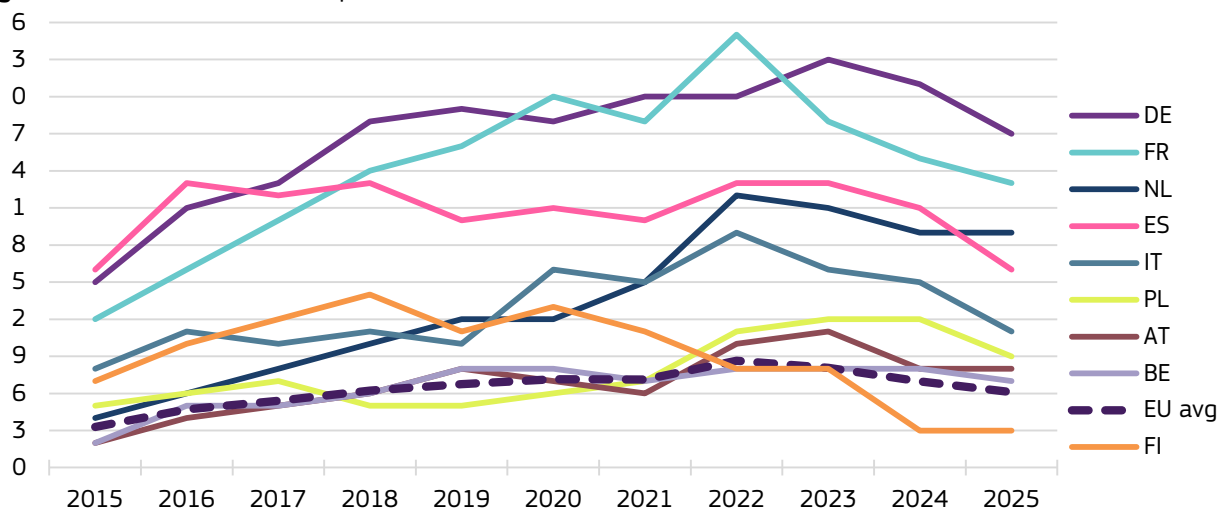
with 10%, followed by Spain at 9%. Coming back to startups, Italy (7%) comes up behind the Netherlands (8%), similar to the distribution of digital firms overall. The top four countries (DE, FR, ES and NL) together take more than half of the share of European VW startups. Finally, the number of VW startups in Estonia (3% of the EU's share) is remarkably high in comparison to the total number of players.

Figure 25 presents the growth in the number of startups for the EU countries with the largest number of them over the period 2015-2025. Overall, the number of VW startups has been growing rapidly over the years, reaching its peak in 2022 with more than 230 startups across the EU.⁵ Only three countries – Finland, Romania and Slovenia – have a smaller number of young firms compared to a decade ago. The case of Finland is particularly significant, a country in the EU’s top-10 in terms of the number of startups. However, since 2015, the Finish VW startup landscape has shrunk 57%. However, only one startup appears in the top 20 players engaged in VW patenting activity: Finnish-based Varjo Technologies Oy, which started as a startup in 2016 in Finland. So while the overall Finnish VW startup environment has shrunk, unicorns that have scaled up exist. Continued investment in startups in Finland may help boost stalled activity in this area. Of note is also that the majority of VW patenting activity is led by Finnish-based Nokia and research by Tampere university based in Finland.

The Netherlands, Austria and Belgium have experienced a significant VW startup boom. The Dutch VW hub is the 4th largest in 2024 –and ranks 3rd in the provisional 2025 data.

Since 2022, the number of startups has seen a downtrend, either because they have scaled up into growth firms or because they are not able to attract further early VC deals.⁶ As for the EU aggregate, the number of startups peaked in 2022 for most Member States, although in some countries the highest number was reached earlier in 2021 (Luxembourg, Denmark) or later in 2023 (Germany, Austria, Poland).

Figure 25. Growth of VW startups in selected EU countries.



Source: JRC DGTES database.

Note: The line chart presents annual indexed numbers over base year (2015) for the EU Member States with the largest number of startups. Data labels on the right reflect the total growth in the number of startups in each EU Member State over the period 2009-2024 (2025 date are yet provisional). ‘EU avg’ refers to the average figures for the EU27 aggregate.

These trends exhibit heterogeneous dynamics in startup development. While some countries might be reducing the number of startups as they are scaling-up, it is important to keep a base of startup development, as these firms bring innovative ideas to the industry. While the EU must promote firm growth to develop a competitive VW industry, it must do this in a sustained way by unlocking access to capital within the EU, to avoid unicorns being acquired by non-EU countries and a few startups being funded by a handful of EU investors. ■

5. Venture capital data in 2025 are only partially available, thus figures for this year should be interpreted with caution. Data prior to 2015 are not representative enough, and it has been left out of this part of the analysis.

6. DGTES assigns the status of startup in a given year to firms receiving at least one early VC deal that year and for the following four years. If, during the period of 4 years, the firm receives another early VC deal, the count restarts for another four years. However, if the firm receives a late stage VC or a private equity deal, it is considered to qualify as growth firm, losing the status of startup for that year and for the eventually remaining years.

04 DEPENDENCIES AND INTERDEPENDENCIES

As virtual worlds and immersive technologies rapidly evolve into foundational components of the future digital economy, understanding the ownership structures that shape this emerging ecosystem has become a strategic imperative for the European Union. Virtual worlds firms—encompassing developers of virtual reality (VR) and extended reality (XR) hardware, software, platforms, engines, content infrastructure, and immersive services—are no longer niche players in the tech landscape. Instead, they are increasingly central to sectors such as education, industry, culture, health, and public administration. As these technologies scale, ownership and control over the firms developing and operating them will profoundly influence economic competitiveness, data governance, security, and the cultural and societal values embedded in digital environments.

The ownership of virtual worlds firms determines who controls the intellectual property, data flows, platform governance, technological standards, and economic rents associated with immersive technologies. Examining both directions of ownership—EU firms investing in or acquiring virtual worlds companies abroad, and non-EU firms buying or investing in EU companies—is therefore crucial. EU ownership of firms overseas can strengthen Europe's global influence, deepen its participation in global value chains, and ensure access to key technologies and talent pools. Conversely, foreign ownership of EU-based firms can bring capital, expertise, and market access, but it can also result in strategic dependencies if critical capabilities migrate outside the Union or if decision-making is ultimately guided by non-European regulatory, commercial, or political priorities.

This dual analysis is especially important in the context of EU debates on digital sovereignty. Digital sovereignty impacts the EU's capacity to shape, regulate, and control its own digital ecosystems, ensuring that critical technologies reflect European values, security requirements, and economic interests. In an environment where virtual worlds are expected to become important infrastructures—hosting everything from industrial simulations to social interaction, commerce, education, and public

services—sovereignty concerns intensify. Ownership concentration by non-European actors could limit the EU's ability to regulate platform architecture, safeguard sensitive data, mandate interoperability, or prevent lock-in to closed ecosystems. At the same time, insufficient levels of European outward investment may weaken the EU's position in global technology governance and reduce its influence over the standards and norms shaping future immersive environments.

Studying foreign ownership trends in virtual worlds firms therefore provides a critical lens through which to assess Europe's long-term technological autonomy and strategic resilience. It offers insights into where value is created and captured, how global competitive dynamics are shifting, and what structural vulnerabilities may emerge as immersive technologies become ubiquitous. Understanding these patterns will be essential for informing investment screening mechanisms, R&D strategies, industrial policy interventions, and broader efforts to secure a robust, competitive, and values-aligned European presence in the rapidly developing virtual worlds ecosystem.

Note that this section reports on firm ownership and excludes government and research organisations. Foreign ownership is defined as ownership of more than 50% of shares (IMF, 2009) by an entity located in a country different from the country of residence of the firm. We further distinguish **foreign**, **domestic**, and **local ownership** from an EU perspective. Therefore, **foreign-owned EU firms** are EU-based firms owned by investors located in countries outside the EU. **Domestic ownership** refers to EU firms that are owned by entities located within the EU (e.g. a French firm owned by a German investor). **Locally-owned** firms are those owned nationally (e.g. an Italian firm that has Italian ownership).

4.1 Foreign Ownership of Global Virtual World Firms

The distribution of foreign-ownership in virtual worlds firms reveals an internationalised ecosystem. Figure 26 below shows that capital

flows in the virtual worlds sector are cross-border, with several countries playing central roles in financing and controlling virtual worlds enterprise activity worldwide. The diagram visualises how foreign owners (left) hold shares in virtual world firms located in various geographic regions (right). Understanding these patterns is critical for anticipating geopolitical dependencies, assessing potential economic security risks, and designing appropriate regulatory responses.

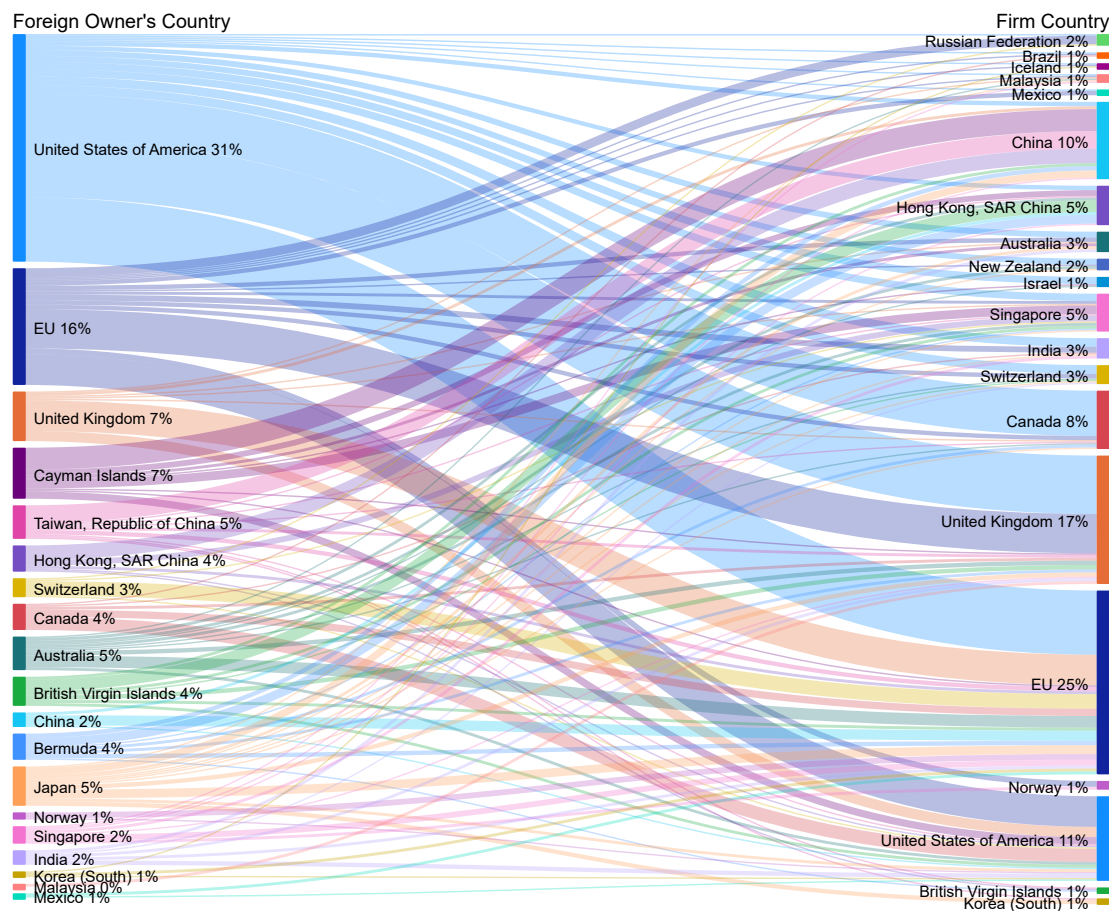
The EU emerges as the second-largest foreign owner of virtual worlds firms (16%). EU investors show a majority of foreign ownership shares in the UK and the US. The EU in turn receives investment from several sources, the largest

being the US, followed by the UK, and including Switzerland, China, and Japan.

The US holds the largest share of foreign-owned firms worldwide (31%), accounting for roughly one-third of all cross-border ownership of virtual worlds firms. US investors are particularly prominent in the UK, EU, and Canada.

The UK plays a dual role as both a foreign investor and one of the world's largest destinations for foreign virtual worlds ownership. Nearly one-fifth of all foreign-owned virtual worlds firms are based in the UK, and investment comes from a diverse array of sources including the US and EU.

Figure 26. Control of foreign virtual world firms 2009-2025.

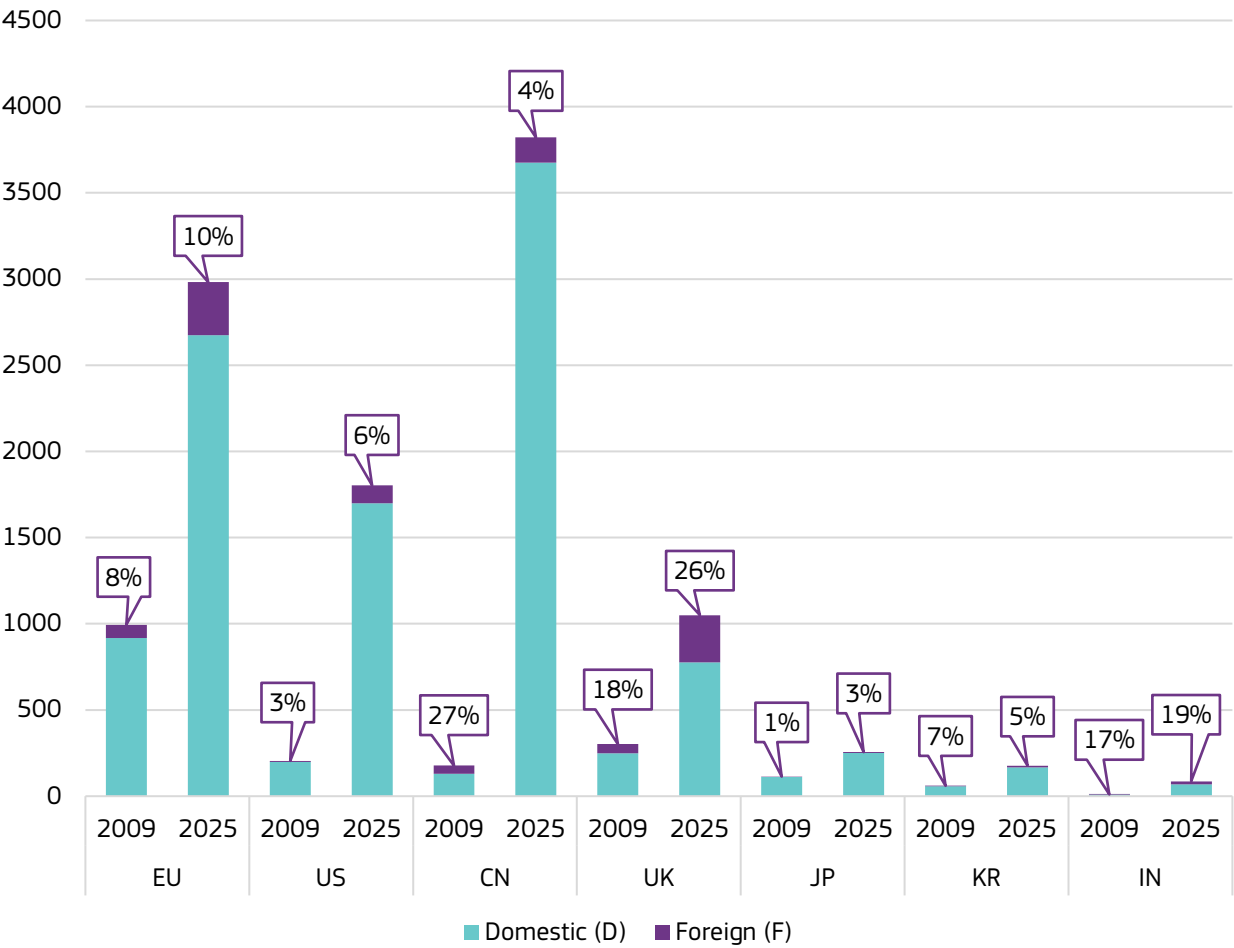


Source: JRC DGTES database.

Note: Countries that own less than 0.5% of globally foreign-owned firms have been excluded from this analysis.

Overall, the figure illustrates that virtual world ownership is not geographically siloed but heavily transnational.

Figure 27. Total numbers and shares of local and foreign-owned firms in selected geographical areas, 2009 vs 2025.



Source: JRC DGTES database.

Note: Geographical areas refer to European Union (EU), United States (US), China (CN), United Kingdom (UK), Japan (JP), South Korea (KR) and India (IN). Data labels present the share of foreign firms over the total number of VW firms for which ownership data is available.

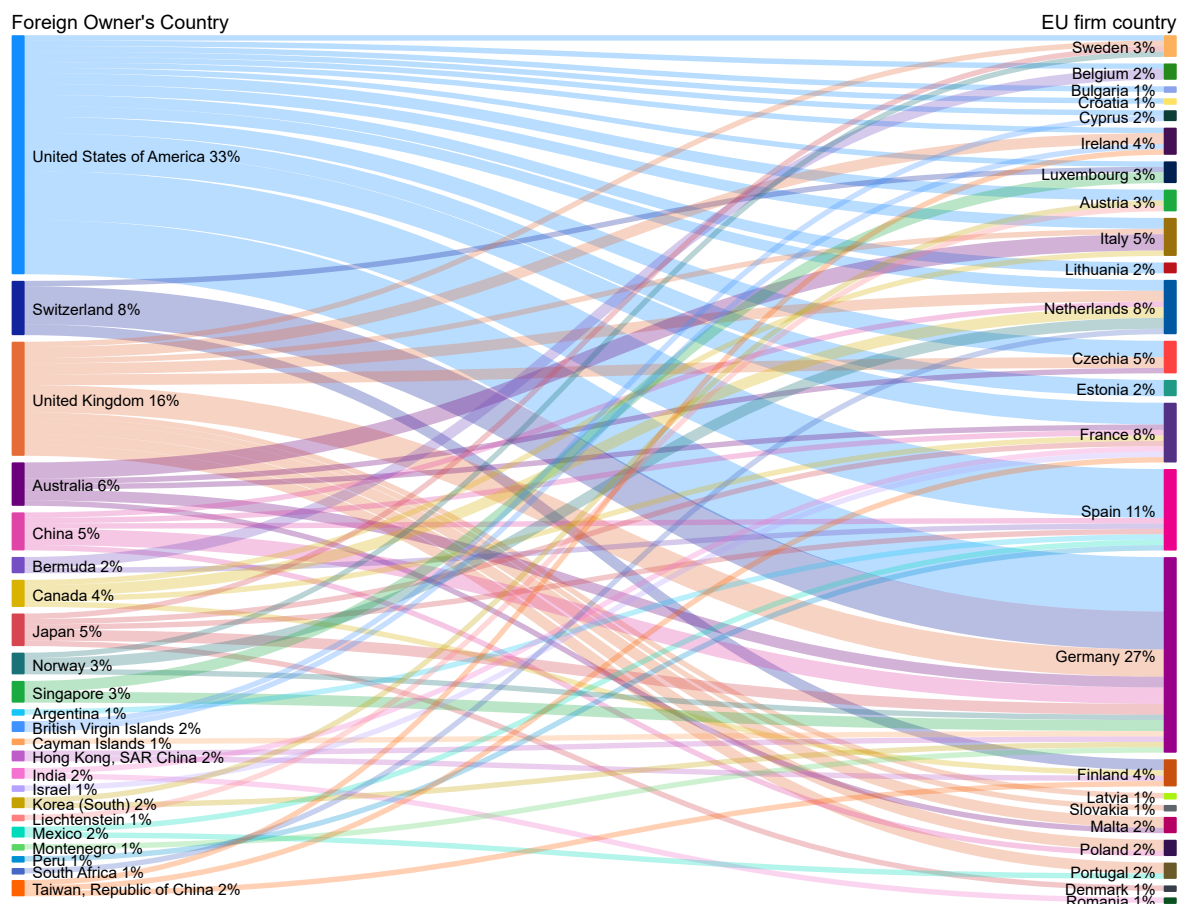
Figure 27 illustrates both foreign and domestic ownership of VW firms between 2009 and 2025 in the EU, US, China, UK, Japan, Korea, and India. The share of foreign firms has significantly decreased in China due to a rapid growth of local businesses, increased markedly in the UK, increased slightly in Japan, the EU, the US and India, and somewhat decreased in South Korea. Of particular note is the growth in domestic firms in China, pushing down the share of foreign firms from 27% in 2009 to just 4% in 2025.

While the EU shows heavy intra-EU investment, similar to the US, EU ownership of foreign firms has expanded between 2009 and 2025.

4.2 Foreign Ownership of EU Virtual World Firms

Figure 28 illustrates how foreign capital flows into virtual worlds firms located in EU Member States, showing which foreign (non-EU) countries own stakes in EU-based virtual worlds firms and how this investment is distributed across the EU.

Figure 28. Foreign ownership of EU virtual world firms 2009-2025.



Source: JRC DGTES database.

The US is the dominant foreign owner of EU virtual worlds firms (33%). UK capital flows into almost all major EU virtual worlds hubs, particularly Germany (27%), Spain (11%), France (8%), and the Netherlands (8%).

The UK holds the second largest share of foreign-owned EU virtual worlds firms at 16%, investing mostly in Germany but also spread across Portugal, Poland, Finland, the Netherlands etc. The UK therefore remains a central financial backer of EU technological innovation, maintaining close integration with the EU's virtual worlds ecosystem. Switzerland is also a major continental investor holding 8% of foreign stakes in EU virtual worlds firms. Switzerland invests broadly across the EU, but particularly in Germany and Finland.

A majority of foreign-owned virtual worlds firms are located in Germany (27%), followed by Spain

(11%), France (8%) and the Netherlands (8%) (Figure 28).

4.3 EU Ownership of Virtual Worlds Firms Abroad

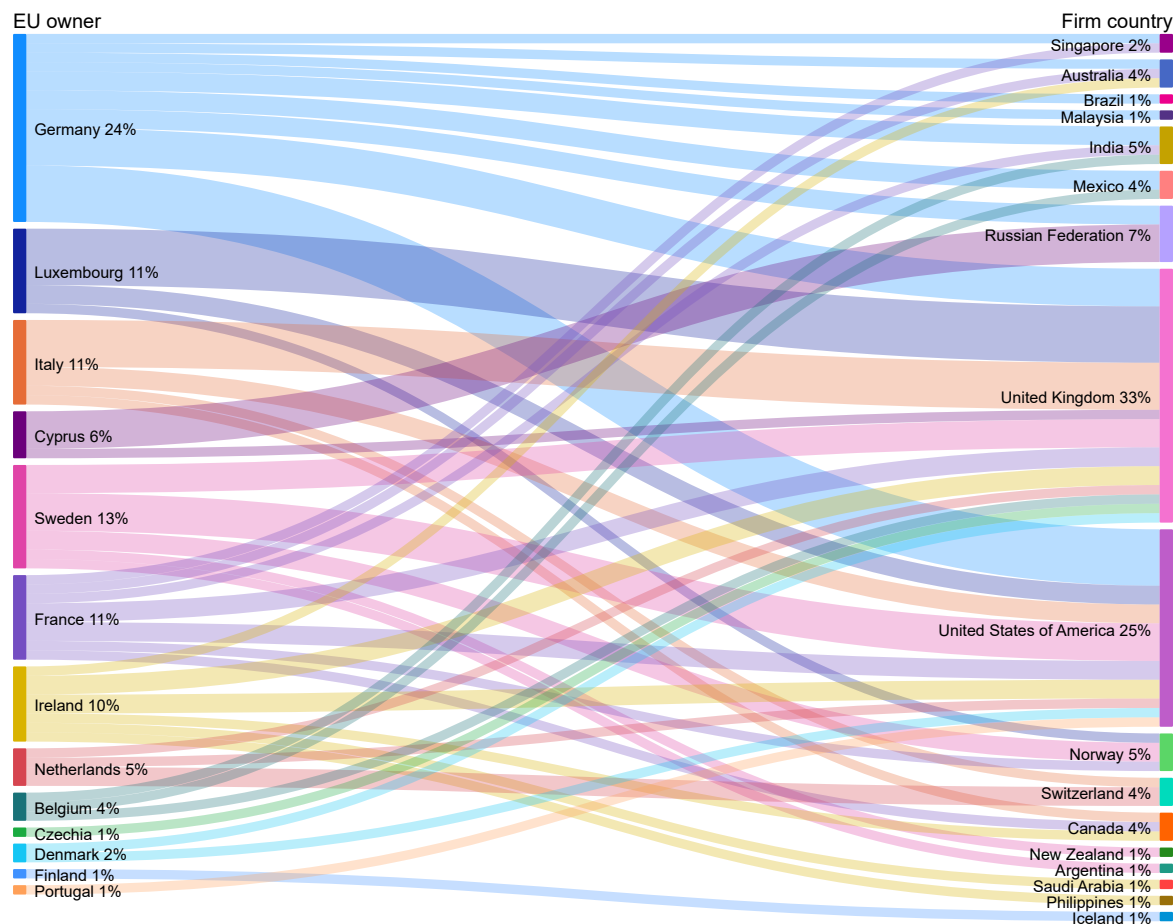
Figure 29 shows how EU-based investors (left) hold ownership stakes in virtual worlds firms located outside of the EU (right). It illustrates how European capital is distributed across the global virtual worlds landscape and which non-EU economies are most closely connected to the EU through investment.

At a high level, the picture shows that a majority of EU investment in foreign destinations is located in the UK (33%) and the US (25%). This reflects deep financial, technological, and corporate linkages between EU economies and the UK and US. A majority of foreign investment is sourced

in Germany (24%), Sweden (13%), France (11%), Luxembourg (11%) and Ireland (10%). While trends in German leadership in virtual worlds foreign ownership mirror their activity leading in

virtual worlds within the EU, Sweden stands out here as a major foreign investor in virtual world activity abroad.

Figure 29. EU ownership of foreign virtual world firms 2009-2025.



Source: JRC DGTES database.

It should be noted that EU capital also flows into a diverse set of additional non-EU markets. Notable destinations include the Russian Federation (7%), Norway (5%), India (5%), Mexico (4%), Australia (4%), Canada (4%), and Switzerland (4%). This shows EU investment in both advanced and emerging markets.

In conclusion, the distribution of EU foreign ownership shows both strategic focus and global diversification. Even though the virtual worlds ecosystem is nascent yet, comprising only 2% of global digital activity, the EU is investing abroad, building strategic ties and regional linkages. While

the UK and US together account for 58% of EU foreign-owned virtual worlds firms, the EU is also showing emerging trends in investment in Asia, Oceania, and other parts of the globe.

In conclusion, this analysis of foreign ownership in virtual worlds firms underscores the deeply interconnected and transnational nature of this emerging digital sector. The data reveals a global ecosystem where capital flows across borders, with the EU, US, and UK playing central roles in shaping the ownership and control of virtual world enterprises. For the EU, this interconnectedness presents both opportunities and challenges. While

its position as the second-largest foreign investor in the sector (16%) highlights its capacity to influence global digital innovation, the dominance of the US (31%) and the UK's dual role as both investor and destination for foreign capital underscore the need for strategic alignment to safeguard economic sovereignty and competitive advantage.

The trends observed over the 2009–2025 period illustrate the evolving dynamics of this sector. The EU's expansion of foreign ownership, particularly in the UK and US, reflects its integration into global technological ecosystems. However, the diversification of EU investments into emerging markets—such as India, Mexico, and the Russian Federation—demonstrates a forward-looking strategy to balance traditional partnerships with new opportunities in Asia, Oceania, and beyond. This dual approach not only strengthens the EU's global economic ties but also mitigates over-reliance on any single region, enhancing resilience in the face of geopolitical and market uncertainties.

China's rapid domestic growth in the virtual worlds sector, which has reduced the share of foreign-owned firms from 27% to 4%, signals a shift in global power dynamics. Similarly, the UK's emergence as the leading destination for foreign-controlled virtual world firms (20%) and its continued integration with EU ecosystems highlight the importance of maintaining transnational cooperation post-Brexit.

For policymakers, the findings underscore the urgency of addressing three critical areas. First, the EU could look to deepen its regulatory frameworks to ensure that foreign investments align with principles of data privacy, ethical AI, and digital sovereignty. Second, fostering domestic innovation and attracting strategic capital will be vital to counterbalance the dominance of U.S. and Chinese players. Third, the EU's growing investments in non-EU markets present opportunities to shape global standards for virtual worlds, particularly in areas like blockchain governance and decentralised identity systems.

In conclusion, the virtual worlds sector is a microcosm of the broader digital economy—characterised by fluid ownership structures, rapid technological evolution, and geopolitical interdependence. By leveraging its strengths in cross-border collaboration, ethical innovation, and strategic diversification, the EU can position itself as a leader in this transformative space. Doing so requires not only a clear understanding of current ownership patterns but also proactive policies to harness the sector's potential for economic growth, technological leadership, and global influence. ■

05 VIRTUAL WORLDS, INDUSTRIAL ECOSYSTEMS, AND KEY ENABLING DIGITAL TECHNOLOGIES

The integration of virtual worlds into industrial ecosystems marks a pivotal shift in how industries innovate, collaborate, and operate in the digital age. Beyond their origins in gaming and entertainment, virtual worlds are now redefining value chains across sectors such as manufacturing, healthcare, education, and logistics. These immersive, technology-driven environments enable real-time simulation, remote collaboration, and data-driven decision-making, offering transformative potential for efficiency, sustainability, and scalability. As industries navigate the complexities of digital transformation, understanding the interplay between virtual worlds and industrial ecosystems becomes critical to unlocking competitive advantages and fostering resilience in a rapidly evolving global economy.

This chapter introduces a comprehensive analytical framework—a matrix to map virtual world activities across 13 industrial ecosystems. By categorising and evaluating how each ecosystem leverages virtual world technologies, the matrix serves as a tool to identify trends, gaps, and opportunities for cross-industry innovation. From smart manufacturing and decentralised supply chains to virtual healthcare and immersive education, the matrix illustrates how virtual worlds are not merely complementary tools but foundational elements of next-generation industrial systems. It also highlights the varying degrees of maturity and adoption across sectors, providing a snapshot of where industries stand in their digital evolution.

Central to this transformation is the role of key enabling technologies, such as XR, AI and the Internet of Things (IoT). AI underpins the functionality of virtual worlds by enabling dynamic environments, predictive analytics, and personalised user experiences. For instance, AI-driven algorithms optimise real-time data processing in virtual simulations, enhance automation in industrial training, and power intelligent avatars in collaborative platforms. Coupled with advancements in blockchain, 5G/6G and the Internet of Things (IoT), AI is

accelerating the integration of virtual worlds into industrial workflows, fostering smarter, more interconnected ecosystems.

This chapter provides an analysis of virtual world activity across 13 industrial ecosystems, as well as the performance of key enabling technologies, namely XR, AI, 5G & Connectivity, IoT, Blockchain, and 3D Printing related to virtual worlds. The subsequent sections first offer granular insights into each of the 13 ecosystems, followed by an in-depth analysis of the technologies, including AI, that are shaping their evolution along the lines of research publications, innovation patenting, and business activity. Through this exploration, the chapter seeks to illuminate pathways for the EU to harness virtual worlds as catalysts for growth, sustainability, and global competitiveness.

5.1 Virtual Worlds Across Industrial Ecosystems

For the purposes of this report, industrial ecosystems are defined as interdependent techno-economic spaces where activities take place in an interconnected way and with the participation of heterogeneous actors pursuing different socioeconomic goals. The EU Industrial Strategy (European Commission, 2021) identified 14 policy relevant industrial ecosystems. Virtual world activity across the thirteen⁷ industrial ecosystems (other than digital) are analysed: Aerospace & Defence, Agrifood, Construction, Cultural and Creative Industries, Electronics, Energy Intensive Industries, Energy-Renewables, Health, Mobility-Transport-Automotive, Proximity, Social Economy and Civil Security, Retail, Textiles, and Tourism.⁸ A fourteenth industrial ecosystem, Digital, is not included in this analysis given the

7. The digital ecosystem is not included here due to the sheer volume virtual world activities that are concentrated within the digital ecosystem that would skew analyses. The virtual world activities in the digital ecosystem are analysed at length in other sections of this report.

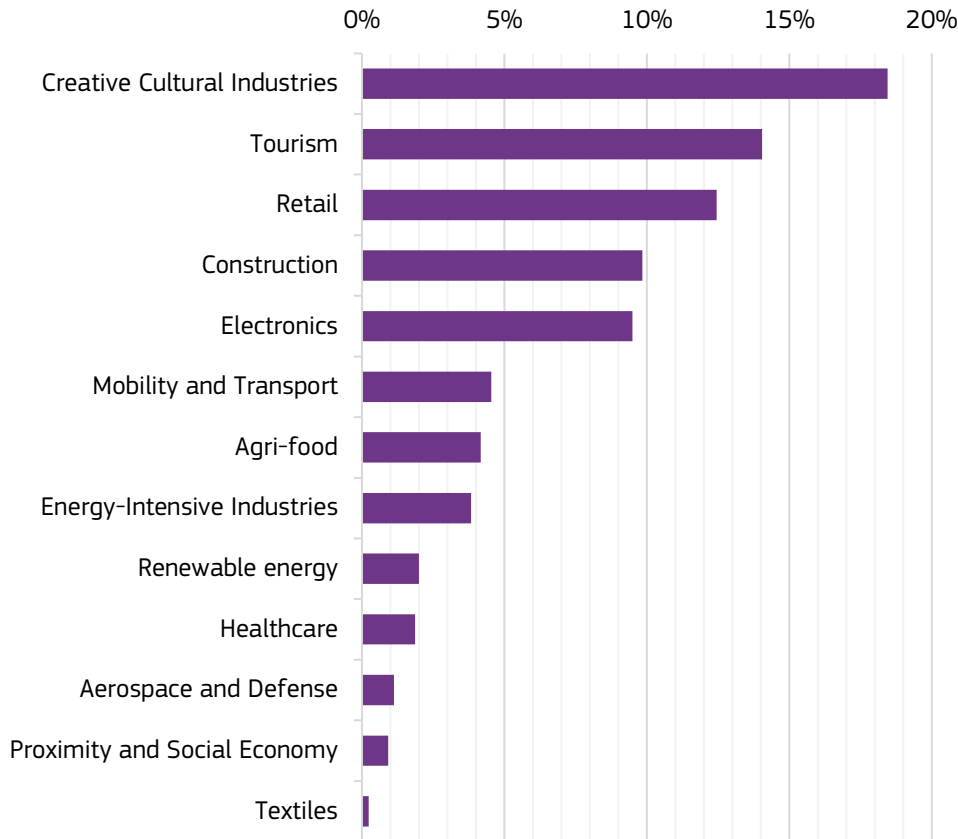
8. Together with a fourteenth industrial ecosystem, Digital, these ecosystems represent approximately 70% of the EU economy and 80% of the business economy (as a share of value added) (COM(2021) 350 final): <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52021DC0350>

sheer number of virtual world activities located within this ecosystem (100% of virtual world activities overlap with the digital ecosystem) that would affect visualisation and analysis.

First, global virtual world activity across industrial ecosystems overall shows that almost 20% of virtual world activity is found in Creative Cultural Industries. Virtual world activity is also found

within Tourism (14%), Retail (13%), Construction (10%), and Electronics (10%). Electronic-related VW activity included those related to VW-hardware (sensors, headsets etc.). Despite possibilities for their applications in industries such as aerospace & defence (currently at 1%) and healthcare (2%), for training purposes for example, use of virtual worlds within these ecosystems seems to be as yet limited.

Figure 30. Global virtual world activity across industrial ecosystems 2009-2025.



Source: JRC DGTES database.

Note: The digital ecosystem is not included here due to the sheer volume virtual world activities that are concentrated within the digital ecosystem that would skew the figure. The virtual world activities in the digital ecosystem are analysed at length in other sections of this report.

The large share of virtual world activity within Creative Cultural Industries is unsurprising, given its natural alignment with immersive technologies used for gaming, digital art galleries, virtual museums, metaverse-based performances, and NFT-driven creative ecosystems. Within the tourism industry, there are examples of virtual worlds to cultivate immersive travel

experiences such as virtual tours of heritage sites to help reduce environmental footprints, as well as experiential storytelling. In retail, virtual worlds can help create virtual shopping experiences, digital fashion showrooms and fitting rooms, and NFT-based product trials. Virtual worlds have proven useful in architecture within the construction ecosystem. However, the

relatively smaller share of virtual world activity in other industrial ecosystems could result in fragmentation and uneven economic benefits. Healthcare, Aerospace & Defence, and Energy-Intensive Industries could benefit from virtual solutions for remote diagnostics, employee training, and flight and defensive and attack simulations and planning. Ecosystems where virtual world activity may be emerging such as Mobility and Transport could see the use of this technology to support smart city designs and sustainability-focused urban planning.

Analysis of the activities of virtual world players (firms, government organisations, and research institutes) in the EU, US, China, UK, and South Korea across these thirteen industrial ecosystems illustrates distinct trends. In the EU, after Creative Cultural Industries, players are most engaged in Tourism (19%), Construction (15%), and Retail (12%), echoing global trends. The EU also shows higher engagement of virtual worlds in Healthcare (7%) as a share of its total activities across industrial ecosystems in comparison to its global counterparts, signalling a foundation for growth and competitiveness in this area.

Figure 31. Virtual world player engagement across industrial ecosystems.

	EU	US	CN	UK	KR
Aerospace and Defence	3%	4%	1%	2%	1%
Textiles	<1%	<1%	<1%	1%	1%
Proximity and Social Economy	2%	3%	<1%	3%	<1%
Healthcare	7%	6%	1%	4%	8%
Agri-food	3%	4%	6%	2%	3%
Renewable Energy	1%	<1%	3%	<1%	2%
Energy Intensive Industries	2%	2%	6%	2%	1%
Mobility and Transport	3%	4%	5%	2%	12%
Retail	12%	13%	16%	11%	5%
Creative Cultural Industries	29%	26%	20%	33%	23%
Tourism	19%	20%	16%	22%	12%
Electronics	3%	5%	14%	3%	7%
Construction	15%	13%	12%	14%	24%

Source: JRC DGTES database.

Note: Deeper hues correspond to higher numbers of players. Shares taken of total number of activities for each country across the 14 industrial ecosystems except Digital, given the sheer number of virtual world related activity located in the Digital industrial ecosystem that would skew figures.⁹ Shares less than 1% are indicated by <1%. These figures are meant to provide an indication of the proportion of each technology in industrial ecosystems and are dependent on the data sources that constitute the DGTES database (see section 1.2).

In China, virtual world engagement in Retail is relatively high (16%), along with Tourism (16%), Electronics (14%) and Construction (12%). China also stands out showing relatively high engagement of virtual world players in Agri-food (6%), Energy-Intensive Industries (6%), Mobility and Transport (5%), as well as Renewable energy (3%), indicating how these industries are increasingly leveraging the benefits of virtual worlds in these ecosystems in China.

9. 66% of virtual world activities are located within the Digital industrial ecosystem in the EU, 62% in the US, 60% in the UK, 37% in China, and 76% in South Korea.

South Korea shows high engagement of virtual worlds in Construction (24%), along with engagement in Creative Cultural Industries at 23%. A majority of UK virtual players are found in Creative Cultural Industries (33%), with others mostly in Tourism (22%), Construction (14%) and Retail (11%). The US sees engagement of virtual worlds in Creative Cultural Industries (20%), followed by Retail (16%), Tourism (16%), Electronics (12%) and Construction (12%).

A deep dive into the share of EU research publications, innovation patenting, and business activities, along with EU funded projects that are related to virtual worlds across industrial ecosystems reveals nascent yet distinct trends. A majority of EU funded projects on virtual worlds are found within Creative Cultural Industries (21%), Construction (20%), and Healthcare (14%).¹⁰ The huge discrepancy between 22% of EU research activity on healthcare related to virtual worlds only translating to 2% of business engagement indicates a missed opportunity to leverage research insights into business gains, perhaps linked to access to capital and siloed research and business environments.

Figure 32. Share of EU research publications, innovation patenting, EU projects and business activities across industrial ecosystems.

	Business	Innovation	Research	EU funded projects
Aerospace and Defence	1%	1%	1%	5%
Textiles	<1%	<1%	<1%	<1%
Proximity and Social Economy	1%	<1%	<1%	3%
Healthcare	2%	5%	22%	14%
Agri-food	5%	2%	3%	4%
Renewable Energy	2%	2%	1%	1%
Energy Intensive Industries	5%	1%	7%	7%
Mobility and Transport	4%	16%	7%	4%
Retail	16%	2%	7%	4%
Creative Cultural Industries	22%	24%	12%	21%
Tourism	18%	3%	4%	9%
Electronics	12%	6%	3%	7%
Construction	10%	36%	33%	20%

Source: JRC DGTES database.

Note: Deeper hues correspond to higher numbers of players. Shares taken of total number of activities for each area (business/research/innovation/EU funded) across the 14 industrial ecosystems except Digital, given the sheer number of virtual world related activity located in the Digital industrial ecosystem that would skew figures. Shares less than 1% are indicated by <1%. These figures are meant to provide an indication of the proportion of each technology in industrial ecosystems and each area (business/research/innovation/EU funded) and are dependent on the data sources that constitute the DGTES database (see section 1.2).

10. When not taking into account the Digital Ecosystem, which would lead across all dimensions with regard to virtual world activity. Note also that percentages in this section are calculated as a whole of activities within each area i.e. research publications, patenting, business activity, and EU funded projects, and not normalised across, given differing amounts of activities within each type of activity i.e. the total number of business activities far outweighing that of innovation patenting or research.

EU research publications related to virtual worlds are mostly found within Construction (33%) and Healthcare (22%), the latter which corresponds to the relatively larger share of EU player engagement in healthcare in comparison to its global counterparts (see Figure 31). EU business activities are mostly concentrated in Creative Cultural Industries (22%), followed by Tourism (18%), Retail (16%), Electronics (12%) and Construction (10%). EU innovation patenting activity is mostly related to Construction (36%), Creative Cultural Industries (24%), and Mobility & Transport (16%).

In summary, Creative Cultural Industries and Construction emerge as industrial ecosystems that see engagement across business, research, and innovation. In the Construction sector, EU-funded activities (20%) and business engagement (10%) show some alignment with research outputs (33%) and innovation patenting activity (33%), suggesting coherence in the innovation-to-market pathway. While this highlights trends in early adoption, it also underscores fragmentation in how research, funded initiatives, and commercial applications intersect across other industries in the EU. Outside of Creative Cultural Industries and Construction, EU business activity is concentrated in Tourism, Retail, Electronics, and Construction, research in Healthcare and innovation patenting in Mobility & Transport.

There is therefore some fragmentation in how virtual world research activities are being translated to business activities or vice versa. For instance, Healthcare—a sector with 22% of EU research activity related to virtual worlds—only translates to 2% of business engagement, indicating a missed opportunity to leverage virtual technologies for applications such as remote diagnostics, patient training, and medical training and simulations. There also seem to be some emerging opportunities for business-related VW activity on Mobility & Transport (4%) to support and be supported by innovation patenting in this area (16%) and research (7%). Similarly, Energy-Intensive Industries show modest business activity (5%) and some gains in research (7%),

pointing to untapped potential for innovation in these sectors.

These findings underscore the need for collaborative innovation within and across industrial ecosystems. For instance, research and innovation within one ecosystem should support and be supported by business activity in the same. Cross-industry partnerships could accelerate adoption and reduce duplication of efforts e.g. a hardware training module developed for A&D training could be utilised in healthcare or even tourism and retail. Finally, certain industries like Aerospace, Agri-food, and Energy-Intensive Industries remain underutilised, despite their capacity to benefit from virtual tools for simulation, sustainability, and operational efficiency.

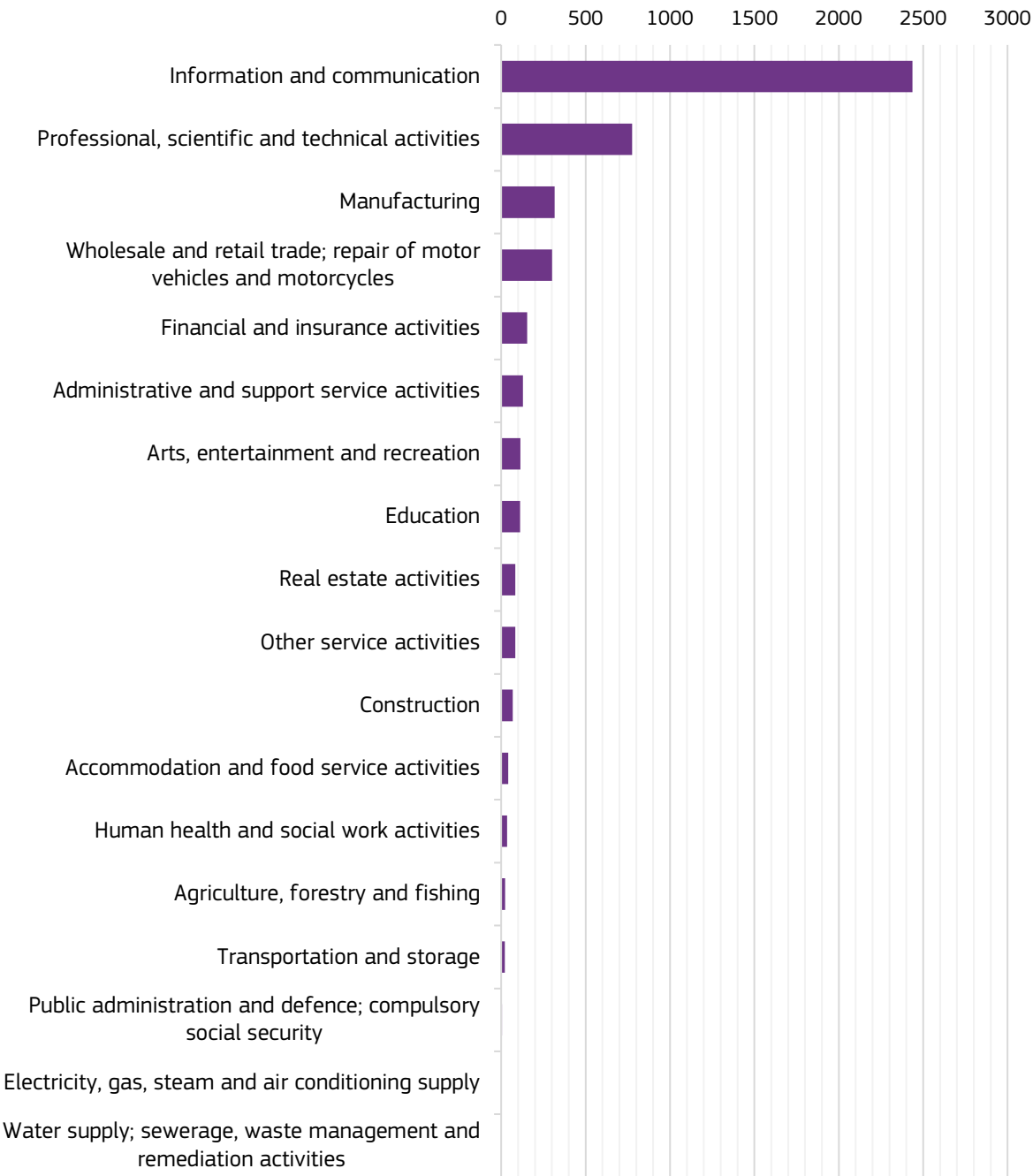
5.2 Virtual Worlds Across Sectors

Sectoral analysis of virtual worlds is distinct from that of industrial ecosystems. The European Commission distinguishes between sectors (classified by NACE codes) and industrial ecosystems as complementary but conceptually distinct frameworks for understanding economic activity and policy design. Sectors are standardised classifications of economic activities defined by the NACE (Nomenclature of Economic Activities) system. NACE codes categorise industries e.g. Manufacturing, Information & Communication. Sectors are defined by predefined boundaries (e.g., ‘manufacturing’, ‘mining & quarrying’, ‘real estate activities’) and NACE codes standardise how economic data (e.g., GDP, employment, trade) is collected, analysed, and reported across the EU. Industrial ecosystems on the other hand are dynamic, interconnected networks of sectors, stakeholders, and technologies working collaboratively to achieve systemic goals. They transcend traditional sectoral boundaries. This section presents both to help address how virtual world activities perform in both industrial ecosystems and sectors, with granular level analysis on the former.

Figures 33 and 34 provide visual comparisons of global and EU trends in virtual world activities across sectors from 2009-2025. In conducting this sectoral analysis, we see that a majority

of virtual world activities are found within information and communication, professional, scientific and technical activities, and retail, with manufacturing coming in fourth.

Figure 33. Global virtual world activities across sectors 2009-2025.

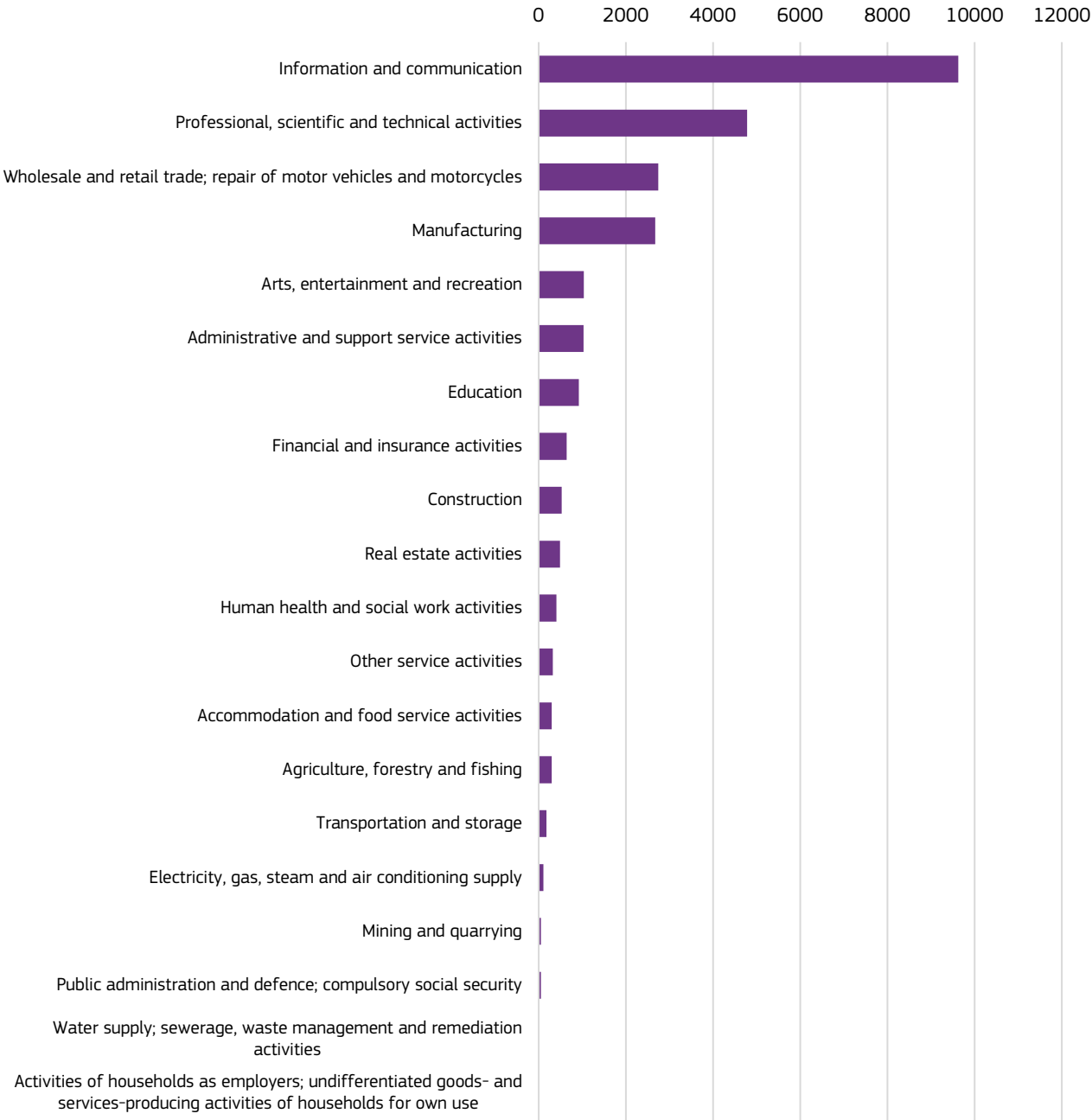


Source: JRC DGES database.

The emphasis on retail activities is complemented when looking at virtual world activities across industrial ecosystems, which show a similar focus on retail-related activities. The concentration of virtual world activities in information and communication (ICT), professional/scientific/technical (PST), and retail sectors highlights their foundational role in enabling, supporting, and monetizing virtual ecosystems. ICT drives the technological infrastructure (e.g., cloud computing, AI, and VR/AR development), while

professional, scientific and technical activities contribute to design, R&D, and consulting services for immersive environments. Retail reflects the demand for virtual goods (e.g., in-game purchases, digital fashion) and e-commerce integration. The inclusion of manufacturing in fourth place underscores its indirect but growing role in producing hardware (e.g., VR headsets, servers) and materials for physical components of virtual infrastructure.

Figure 34. EU virtual world activities across sectors 2009-2025.



Source: JRC DGTES database.

This analysis of virtual world activities across sectors in the EU shows an increased number of activities in the manufacturing sphere, as well as financial and insurance activities. There are also emerging activities related to education, administrative and support services, and retail, in line with global virtual world activities across sectors.

5.3 Key Digital Technologies Enabling Virtual Worlds

At their core, virtual worlds are powered by a convergence of technologies that enable dynamic, scalable, and user-centric experiences. These technologies not only define the functionality of virtual worlds but also drive their potential to transform industries and redefine human-machine interaction. Understanding how these technologies intersect is critical to grasping the future trajectory of virtual worlds and their role in shaping the digital economy.

Extended Reality (XR), encompassing virtual reality (VR) and augmented reality (AR), provides the hardware and software infrastructure for immersive user experiences. They are at the core of virtual worlds, as demonstrated in Figure 35. VR headsets and AR glasses bridge the physical and digital, enabling users to interact with virtual environments in real time. In industrial contexts, AR overlays digital information onto physical workspaces, enhancing remote collaboration or equipment maintenance. Meanwhile, VR facilitates high-fidelity simulations for sectors like healthcare (e.g., surgical training) or education (e.g., virtual classrooms). The integration of haptic feedback and eye-tracking technologies further enhances realism, making virtual interactions indistinguishable from physical ones.

AI is a foundational enabler, driving intelligent automation, real-time decision-making, and adaptive experiences within virtual environments. AI algorithms power procedural content generation, allowing virtual worlds to dynamically create landscapes, characters, and narratives tailored to user behaviour. For example, AI-driven avatars can simulate human-like interactions,

enhancing social and professional collaboration in virtual meetings or training simulations. In industrial applications, AI optimises virtual simulations by analysing real-time data to improve predictive maintenance or supply chain modelling. Additionally, machine learning personalises user experiences, such as adaptive difficulty levels in virtual training or AI-generated recommendations in digital marketplaces.

Blockchain and decentralised technologies underpin the ownership, security, and governance of virtual worlds. Blockchain enables tokenised digital assets, allowing users to own, trade, and monetise virtual goods (e.g., NFT-based land in metaverse platforms) securely and transparently. Smart contracts automate transactions and governance rules, fostering decentralised economies where users retain control over their data and assets. For instance, blockchain-based identity systems ensure privacy and interoperability across platforms, while decentralised autonomous organisations (DAOs) empower communities to co-govern virtual spaces. These innovations align with the EU's push for digital sovereignty, reducing reliance on centralised platforms and ensuring ethical data practices.

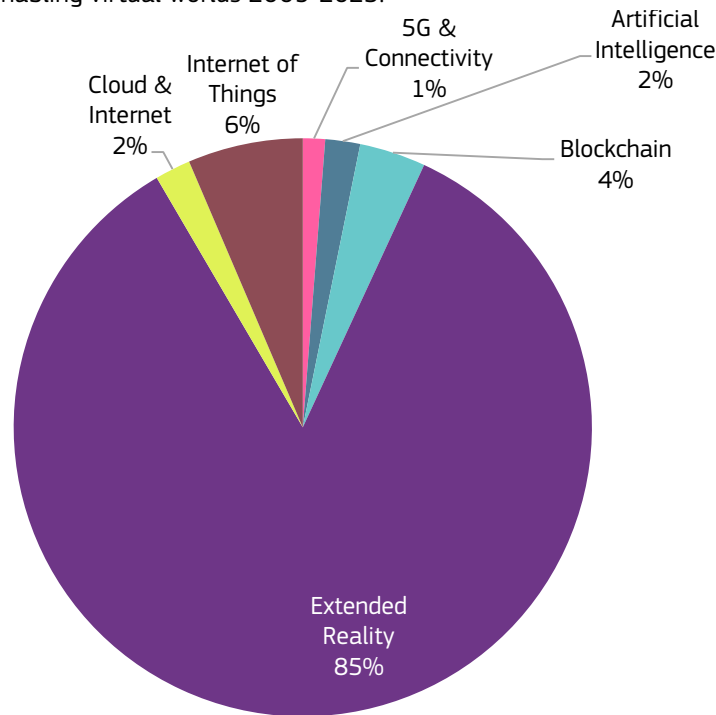
The Internet of Things (IoT) connects virtual worlds to the physical environment, enabling real-time data exchange and environmental responsiveness. IoT sensors feed real-world metrics into virtual simulations, such as monitoring energy usage in smart factories or tracking user biometrics for health analytics. This integration allows virtual worlds to reflect and adapt to real-time conditions, creating hybrid environments where digital and physical systems coexist. For example, a virtual city model might use IoT data to simulate traffic patterns or environmental changes, aiding urban planning and sustainability efforts.

5G, 6G, and edge computing ensure low-latency, high-speed connectivity, which is essential for seamless virtual world interactions. These technologies reduce lag in real-time multiplayer

environments, support high-resolution graphics streaming, and enable distributed computing for resource-intensive simulations. As connectivity expands, it will unlock massively multiplayer

virtual worlds with thousands of concurrent users, fostering global collaboration and economic activity.

Figure 35. Key technologies enabling virtual worlds 2009-2025.



Source: JRC DGTES database.

Note: Digital areas with fewer than 0.5% activities in virtual worlds within the digital ecosystem are not visualised here i.e. 3D Printing.

Figure 35 presents different technologies that comprise VW activity. It highlights the current technological landscape shaping virtual worlds, with Extended Reality (XR) dominating at 85%, underscoring its role as the foundational pillar of immersive digital environments. This overwhelming share reflects XR's integration of augmented reality (AR), virtual reality (VR), and mixed reality (MR) to create interactive, lifelike experiences, from gaming and remote collaboration to virtual training.

Supporting technologies, also called key enabling technologies, though smaller in share, play critical complementary roles. The Internet of Things (IoT) digital area is key in enabling virtual worlds. IoT devices contribute by bridging physical and virtual realms. Sensors, wearables, and smart objects can feed real-time data into virtual worlds (e.g.,

biometric feedback for personalised experiences or environmental monitoring for dynamic simulations).

Blockchain's (4%) inclusion reflects its growing relevance for securing digital ownership (e.g., NFTs), decentralised identity management, and transparent transactions within virtual economies. However, its limited share suggests adoption is still niche, constrained by scalability and regulatory uncertainties.

While modest in percentage, Online Platforms, Cloud & Internet (2%) are vital for hosting, scaling, and delivering virtual world data. Their low representation may reflect their status as enabling utilities rather than visible innovations, yet they underpin real-time rendering, cross-platform access, and low-latency interactions.

AI's role (2%) in virtual worlds—such as generating lifelike NPCs, personalising content, or optimising resource allocation—may yet be growing. The current low percentage may signal current reliance on pre-programmed systems rather than adaptive AI.

The nascent share of 5G & Connectivity (1%) may signal an early adoption phase. As next-gen networks mature, they could enable seamless, low-latency access to virtual worlds, particularly for mobile users and large-scale multiplayer environments.

Figure 36. Activities of key digital technologies enabling virtual worlds in selected geographical areas 2009-2025.

	Extended Reality	Internet of Things	Blockchain	Artificial Intelligence	Online Platforms, Internet & Cloud	5G & Connectivity
EU	88%	3%	3%	2%	4%	<1%
China	82%	11%	3%	2%	<1%	2%
US	86%	2%	4%	3%	5%	<1%
UK	87%	2%	5%	2%	3%	<1%
South Korea	93%	2%	2%	2%	<1%	1%

Source: JRC DGTES database.

Note: Deeper hues correspond to higher numbers of players. Shares taken of total number of activities for each geographic area across the key enabling technologies listed above. Shares less than 1% are indicated by <1%. These figures are meant to provide an indication of the proportion that each technology constitutes each geographic area and are dependent on the data sources that constitute the DGTES database (see section 1.2). China leads in the total number of activities, followed by the US, EU, South Korea, and the UK.

Figure 36 reveals the distribution of activities across key digital technologies enabling virtual worlds in selected geographies. China leads in VW activities related to XR, IoT, Blockchain, AI, and 5G & Connectivity. The US leads in Online Platforms, Internet & Cloud, underscoring its dominance in

digital infrastructure and global digital platforms. The EU holds the third-highest share of activities.

Figure 37 provides a granular view of the top six EU countries most active in the key technologies enabling virtual worlds.

Figure 37. Activities of key digital technologies enabling virtual worlds in selected EU countries 2009-2025.

	Extended Reality	Internet of Things	Blockchain	Artificial Intelligence	Online Platforms, Internet & Cloud
DE	91%	2%	2%	2%	3%
ES	86%	3%	4%	2%	5%
FR	87%	3%	3%	1%	6%
IT	83%	4%	5%	3%	5%
NL	87%	2%	5%	2%	4%
SE	91%	3%	2%	2%	2%

Source: JRC DGTES database.

Note: Deeper hues correspond to higher numbers of activities. Only the technologies that the EU is most active in are analysed above. 5G & Connectivity, in which the EU has a relatively low share of activities as a whole with regard to the virtual world domain, has been excluded. Similarly, given the relative size of the virtual world ecosystem, only the EU countries with the highest number of activities in each of the enabling digital technologies are presented above. Shares taken of total number of activities for each country across the key enabling technologies listed above. Shares less than 1% are indicated by <1%. These figures are meant to provide an indication of the proportion of each technology in each country and are dependent on the data sources that constitute the DGTES database (see section 1.2).

Echoing findings in chapter 1, Germany holds the largest share of activities overall (20%) primarily driven by its leadership in AI (highest share) and XR (also highest share) activities enabling virtual worlds. France follows with 14% of activities, excelling in online platforms, internet, and cloud infrastructure. Italy's 10% share is related to blockchain and IoT, areas critical for securing digital ownership and integrating physical-world data into virtual realms. Spain contributes 10% of activities, with a relatively balanced profile. The Netherlands and Sweden each account for 8% of activities overall across AI, XR, blockchain, online platforms, internet & cloud and IoT, indicating emerging opportunities for virtual worlds.

Figure 38 below provides an analysis of EU, US, and Chinese performance across the three dimensions studied in depth with the DGTES methodology: research publications, innovation patenting, and business activity—in the key enabling technologies for virtual worlds.

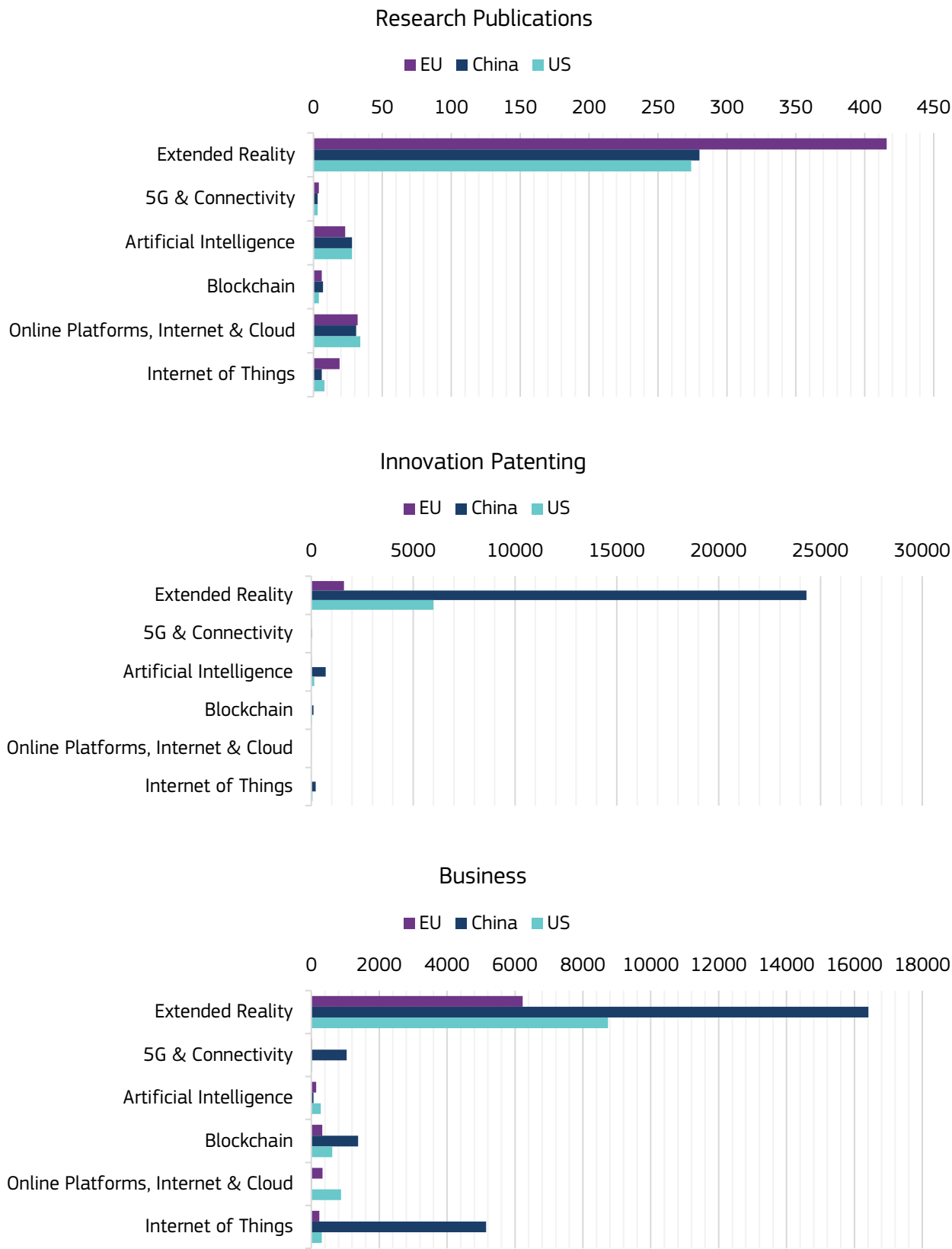
The EU shows leadership in research publications overall, particularly in XR and IoT, foundational technologies for virtual worlds. However, the EU lags behind the US and China in research on AI as it relates to virtual worlds. While XR encompasses the majority of publications overall, Blockchain and 5G & Connectivity show minimal academic activity across all geographic areas, highlighting underdeveloped areas for future investment.

Over 90% of priority patent filings related to virtual worlds are concentrated in XR. The EU trails China and the US in innovation patenting, suggesting a disconnect between research outputs and their translation into industrial applications. The EU may benefit from strengthening innovation-to-market pipelines to convert academic strengths into competitive products.

Business activity in key enabling technologies for virtual worlds is heavily concentrated in XR and IoT, with smaller shares in 5G & Connectivity, AI, and Online Platforms, Internet & Cloud. The underdeveloped status of 5G & Connectivity, AI, and Online Platforms, Internet & Cloud may be areas for the EU to build on in the future. Interest in Blockchain looks to be emerging, and the EU could leverage its foundations in this technology to support future growth.

The EU's leadership in academic research is a long-term asset. Paired with strategies to accelerate innovation-to-market transitions, such as public-private partnerships, improving access to venture capital and aligning academic priorities with industrial needs, business and innovation patenting activities related to virtual worlds could flourish.

Figure 38. Research publications, priority patenting filings, and business activity in each of the enabling technologies for virtual worlds in the EU, China and US 2009-2025.



Source: JRC DGTES database.

Taken together, despite the growing size of the virtual worlds landscape across the EU, trends in specialisation emerge. Germany and France dominate core technologies such as AI, XR, Online Platforms, Internet & Cloud, while Italy excels in niche but transformative areas (Blockchain, IoT). Looking ahead, Germany's AI and XR capabilities could integrate with France's cloud infrastructure to enable large-scale, AI-driven virtual platforms. Italy's blockchain expertise could be vital to enhance trust and data integrity in virtual economies, also in Germany and France. Cross-border partnerships could address fragmentation, such as leveraging Italy's IoT for Germany's industrial XR applications. In terms of the mix of key enabling technologies, across all countries, XR dominates while the rest is spread across comprising 1-5% respectively.

In summary, this section has provided an overview of how seven key technologies (XR, AI, 5G & Connectivity, Blockchain, Online Platforms, Internet & Cloud, IoT, and to a lesser extent, 3D Printing), enable the virtual worlds ecosystem. Extended Reality (XR) dominates the ecosystem, accounting for 85% of activities, with its integration of immersive technologies such as AR, VR, and MR serving as the bedrock for virtual world development. Supporting technologies—including IoT, Blockchain, AI, 5G, and cloud infrastructure—play critical roles in enabling seamless, secure, and scalable virtual environments. However, disparities in adoption and investment across regions and sectors reveal opportunities for targeted policy action.

While the EU leads in research publications on foundational technologies like XR and IoT, its weaker performance in innovation patenting and business activity highlighting opportunities for growth in these domains. By fostering a cohesive ecosystem of emerging technologies, the EU can position itself as a leader in the next generation of virtual worlds—ones that prioritise sustainability, inclusivity, and user empowerment. Ultimately, the evolution of virtual worlds hinges on these enablers, and understanding their dynamics is key to unlocking their transformative potential. ■

06 CONCLUSIONS

The global virtual worlds ecosystem, spanning over 88,000 activities and 68,000 players, is a dynamic and rapidly evolving domain with transformative potential for industries, economies, and societies. This report has mapped the evolution and state of virtual world activities, highlighting the uneven but accelerating adoption of immersive technologies across industrial ecosystems, regions, and sectors. While the EU demonstrates strong research capabilities, it lags in translating these strengths into commercial innovation and business activity compared to the US and China. The concentration of virtual world activity in sectors such as Creative Cultural Industries, Tourism, and Retail underscores their natural alignment with immersive technologies but also reveals fragmentation and missed opportunities in other critical areas, such as Healthcare, Aerospace, and Energy-Intensive Industries.

The EU is a leader in research publications on virtual worlds. However, it trails in patenting and business activity, reflecting the need to bridge academic research and industrial application. The dominance of the US and China in VC funding and patenting patterns highlights the need for the EU to strengthen its innovation-to-market pipelines, foster cross-sector collaboration, and strategically position itself in the global virtual worlds landscape. The report also underscores the transnational nature of virtual world ownership, with the EU's growing investments abroad and its role as a destination for foreign capital. However, ensuring digital sovereignty, ethical innovation, and equitable economic benefits remain pressing challenges.

The interplay of key enabling technologies—XR, AI, 5G, IoT, and Blockchain—forms the backbone of virtual worlds, yet disparities in adoption and investment persist. For the EU, leveraging its research strengths in XR, IoT, and Blockchain, while addressing gaps in AI and 5G & Connectivity is critical to maintaining global competitiveness. The findings call for a coordinated approach to policy, funding, and industrial strategy to ensure the EU can harness the full potential of virtual worlds while aligning with its strategic priorities: digital sovereignty, sustainability, and inclusive growth.

6.1 Summary of Key Findings

GLOBAL AND EU ACTIVITY DISTRIBUTION

Virtual world activities account for 2% of global digital ecosystem activities, with 88,000 activities and 68,000 players (firms, research institutions, and government entities).

China leads with 46% of players and 45% of activities, followed by the US (16% of players, 18% of activities) and the EU (11% of players, 9% of activities).

Within the EU, a majority of activities are carried out in Germany (20%), followed by France (14%), and Spain (10%), with Germany, Italy, Spain, and France leading in business and research activity.

The EU leads in research publications (30% to the US' 16% and China's 14%) but lags in patenting (only just over 1500 of more than 42,000 global filings) and business activity. With regard to patent activity, future research could look into assessing the quality of EU patents versus other geographic countries to assess potential areas for strategic development.

China and the US dominate patenting and business activity, with China's EUR 3 billion in VC funding and the United States' EUR 16.8 billion.

EU PUBLIC FUNDING

Over 3000 business, research and innovation players and over 800 projects have benefitted from EU funded support on virtual worlds between 2009–2025. Within the EU, Germany, Italy, Spain and France are the most active in EU-funded initiatives. Relative to its number of virtual worlds players and activities, Greece hosts a significant number of EU-funded virtual worlds players and projects, coming up behind France on both accounts.

VENTURE CAPITAL FUNDING

With EUR 16.8 billion in funds received over the period from 2009 to 2024, data on VC funding

rounds demonstrate the US's ability to attract high-risk, high-reward investments. China follows, but in comparison, captures just below EUR 3 billion. Notably, the US and China are also ahead in scale and late-stage investments. The EU has raised EUR 1 billion over the same period, with a relatively balanced composition of seed (23%), early (30%) and late-stage (44%) funding. The remaining 2% of European VC funds comes from private equity funds.

OWNERSHIP AND GEOPOLITICAL DYNAMICS

The US is the largest foreign owner of virtual world firms (31%), with significant investments in the UK and EU.

The EU is the second-largest foreign investor (16%), with Germany, Sweden, and France leading investments in the UK and US.

China's rapid domestic growth has reduced foreign ownership from 27% in 2009 to 4% in 2025, signalling a shift in global power dynamics.

KEY TECHNOLOGICAL ENABLERS & INDUSTRIAL ECOSYSTEMS

XR (85% of activities) is the cornerstone of virtual worlds, with IoT, Blockchain, and AI playing complementary but underdeveloped roles.

AI and 5G receive less attention in EU research and business activity related to virtual worlds, despite their critical role in driving scalability and intelligence in virtual environments.

Creative Cultural Industries dominate virtual world activity (20% globally, 33% in the EU), followed by Tourism (14% globally, 27% in the EU) and Retail (13% globally, 24% in the EU).

Healthcare, Aerospace & Defence, and Energy-Intensive Industries show low adoption despite high potential for applications like remote diagnostics, training simulations, and sustainability planning.

The EU's engagement in Healthcare (4%) is relatively higher than its global counterparts, but only 3% of this translates into business activity.

6.2 Policy Implications

Virtual world technologies can bring substantial economic gains. They have evolved from niche gaming spaces into broad digital domains affecting economic governance and competitiveness. For example, digital twins—the virtual representation of real objects—have the capacity to optimise manufacturing and automotive production cycles, redesign industrial processes and reduce waste. The analysis of global and EU activity in virtual worlds reveals a striking contrast between Europe's research strength and its comparatively modest commercial output. While the EU accounts for roughly one-third of worldwide scholarly publications on virtual-world technologies, it contributes only a few percent of the global patent pool and lags behind the US and China in venture-capital-driven business formation. To turn the EU's ideas-rich environment into a market-leading sector, a coordinated set of policy measures is required.

Europe's ability to generate high-impact research must be matched by mechanisms that translate those findings into protectable intellectual property to help close the research-to-patenting gap. Embedding IP counsel within EU funded research clusters—particularly in Germany, Italy, Spain, and France, where business and research activity is already concentrated—could give researchers early guidance on protection strategies and reduce the risk of missed commercial opportunities. Finally, the EU could also link outcomes to market-readiness incentives.

Venture capital support should be rebalanced. The current venture-capital landscape shows the United States and China dominating large-scale, late-stage financing, while EU capital is spread thinly across the seed, early, and late stages. This dispersion hampers the ability of European start-ups to reach the scale necessary for global competition.

Virtual worlds are built on a blend of key enabling technologies. Extended reality (XR) already accounts for the overwhelming majority of activity in virtual worlds, but artificial intelligence, 5G, the Internet of Things, and blockchain remain under-exploited. The EU could look to fund the integration of these complementary technologies into XR solutions. Dedicated integration grants should be earmarked for projects that embed AI-driven agents, edge-AI processing, or low-latency 5G connectivity into immersive environments. A common blockchain interoperability framework will also lower entry barriers for SMEs wishing to develop secure virtual economies.

Cultural and creative industries dominate the current virtual-world landscape, accounting for a third of EU activity. Yet industrial ecosystems with high transformative potential—healthcare, aerospace and defence, and energy-intensive industries—remain largely untapped. The EU should also look to addressing the gap between research and commercial activity in healthcare—where only three percent of research output translates into business.

In terms of ownership and geopolitical control, the US now owns the largest share of foreign virtual-world firms, while the EU is the second-largest foreign investor. China's rapid domestic growth has sharply reduced its foreign ownership, signalling a shift toward self-reliance. To safeguard strategic autonomy, the EU should pay close attention to 'critical virtual-world assets'. Finally, the EU must strengthen supply-chain resilience by mapping dependencies on hardware such as XR headsets and 5G chips, and by incentivising the development of European-based production capacities or diversified sourcing strategies that reduce reliance on any single external supplier.

In conclusion, Europe's research leadership and smattering of unicorns in virtual worlds provides a foundation, but without a concerted effort to bridge the gap to patents, venture capital, and market adoption, the continent risks being

left behind by the United States and China. By aligning public funding with private investment, fostering the integration of AI, 5G, IoT, and blockchain, expanding the sectoral reach beyond cultural industries, and protecting strategic assets through ownership safeguards and supply-chain resilience, the EU can transform its virtual-world ecosystem into a globally competitive, innovative, and strategically autonomous sector. ■

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LIST OF ABBREVIATIONS AND DEFINITIONS

Abbreviations	Definitions
AI	Artificial Intelligence
AR	Augmented Reality
DGTES	Digital Techno-Economic Ecosystem
EU	European Union
MR	Mixed Reality
XR	Extended Reality
VC	Venture Capital
VR	Virtual Reality

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ANNEXES

Annex 1. Virtual Worlds Keywords

(implanted or wearable) medical devices	facial recognition	motion controller	speech synthesis
360 degree view	fantasy	motion input device	strong interaction
3d computer graphics	film	motion recognition	superstruct
3d graphics	first person	motion rendering	teleport
3d human reconstruction	game	movie	teleportation
3d modeling	game controller	mr	telepresence
3d-2d image registration	gamefi	multi-agent optimisation	telework
5g	gamification	multimodal content representation	terraforming
5g connectivity	gaming	multimodal entity expansion	therapy
additive manufacturing	gesture	multimodal inference	tokenomics
advertisement	gesture recognition	multimodal interaction	training simulation
afterlife	glasses	multimodal entity linking	transhuman
afterworld	graviton	multiplatform	transhumanism
agent persona modelling	habitat	multiplayer	transitory metaverse
altspace vr	hand-based input device	multisensory interface	travel
anti-aliasing	haptic feedback	multi-task interaction	travel planning
ar	haptic feedback device	multiverse	tv shows
ar display	harassment	mundivagant	unreality
architecture	head-mounted display	neocosmic	urban area
artificial environment	headsets	neural implicit representation	usability testing
artificial intelligence	healthcare	neurotechnology	user's senses
augmented cognition	holographic	nexus	virtual 3d asset
augmented reality	holographic projection	nft	virtual advertising
avatar	hololens	non-fungible token	virtual alerts

bci	human-computer interaction	object generation	virtual art
bio-inspired sensor	human-machine interaction	object recognition	virtual banking
blended reality	humanoid	omniverse	virtual blogs
blockchain	hybrid reality	online community	virtual business
blockchain technology	hyperreality	online learning	virtual chat room
blogging	hyperspace	online multiplayer	virtual cinema
blogosphere	image generation	online shopping	virtual class
brain-computer interface	image recognition	oral health	virtual coaching
bullying	imaginary place	otherworld	virtual conference
chaingaming	immanency	overlay technology	virtual conferencing
chat room	immersion	pachinko	virtual consulting
cloud computing	immersive	panopticon	virtual counseling
cognition-augmented	immersive device	persistent universe	virtual discussion board
cognitive enhancement	immersive experience	playstation vr	virtual economy
computer-controlled machinery	immersive multimedia	portlet	virtual education
computer-generated imagery	immersive technologies	privacy	virtual embodiment
connected devices	immersive virtual world	privacy-enhancing technology	virtual entertainment
connected sensors	industrial iot	privacy-preserving architecture	virtual environment
controller	industry 4.0	private edge cloud system	virtual event
cosmoplastic	information superhighway	product launch	virtual fashion
cosmos	infosphere	real-estate	virtual forum
crypto	integration optimisation	reality capture	virtual gallery
cryptocurrency	intelligent architecture	realm	virtual good
cyber world	interactive filter	real-time rendering	virtual information
cyber-bullying	interactive overlay	real-world enhancement	virtual landscape
cyberculture	interactive story	real-world interaction	virtual learning
cyber-physical systems	interactive storytelling	remote assistance	virtual marketing

cyberpunk	interactivity	remote sensors	virtual mental health resource
cyberspace	interconnected world	rl based approach	virtual museum
cyborg	intermundane	rl-based approach	virtual object
decentralised finance	internet of everything	robot-assisted surgery	virtual property
deep learning	internet of sensing	safety briefing	virtual reality
defi	internet of things	sandbox	virtual self
dental healthcare	interoperability	scenario evaluation	virtual service
digital fashion	interplanetary	scenario generation	virtual shopping
digital item	iris scanner	scenario population	virtual socializing
digital platform	jabber	scene generation	virtual tool
digitalizing cultural heritage	keyboard	scene recognition	virtual tour
disaster risk management	larp	screen and display technologies	virtual tourism
disaster risk planning	linden	second life	virtual training
disembodiment	live action role playing game	sensors	virtual try-on
distributed sensor networks	machine learning	sim	virtual tutorial
distributed sensors system	machine learning technique	simulation	virtual tutoring
dynamic lighting	machinima	singularity	virtual world
e-business	magic leap	smalltalk	virtual-physical fusion
e-commerce	manage anxiety	smart contract	visor
edge cloud	manage pain	smart devices	vive
education	marker-based tracking	smart glasses	voice recognition
e-government	markerless tracking	smart home	vr
e-learning	marketing	social interaction	vr experiment
embedded sensors and wireless systems	marketplace	social media	vrchat
embedded systems	massively multiplayer online game	social space	vr-enabled banking
embodied interaction	mental health	social virtual world	wearable technology
emotional intelligence	meta	software agent	web 3.0
enhanced reality	metaverse application	sound recognition	web 4.0

entertainment	metaverse-enabled banking	sound synthesis	web of things
escapism	microblog	spacetime	well-being
ethics	microblogging	spatial audio	wireless sensor technologies
extended reality	middleware	spatial computing	working group
eye-tracking	mixed reality	spatio-temporal	workspace
facebook	mmorpg	spatiotemporal data inference	xr
facebook spaces	mobile application	spectacles	
facial analysis	mobile apps	speech recognition	

Annex 2. Observations

REVIEW TOOLS

Please note that the GPT@JRC tool was used to enhance the clarity and coherence of this report in select instances, while ensuring alignment with the European Commission's standards for professional communication.

NON-FRACTIONAL VS. FRACTIONAL COUNTING

Whenever an identified player participates in more than one activity e.g. in venture capital deals or innovation patenting, counting has been done on a **non-fractional** basis (counting them entirely under each of the involved countries and not as a fraction of their participation).

VISUALISATION OF PERCENTAGES

Percentages less than 1% have been excluded from figures for simplicity.

Annex 3. DGTES Digital Area Definitions

This report makes reference to several digital areas which together comprise Virtual Worlds. Extended Reality makes up the largest technology that enables Virtual Worlds, as discussed at length in the report, along with others such as AI, 5G & Connectivity, Blockchain, the Internet of Things (IoT) and 3D Printing. These technologies are defined within the scope of the Digital Techno-Economic Ecosystem Approach (DGTES) as follows:

Extended Reality (XR) encompasses a range of immersive technologies, such as virtual reality and augmented mixed reality. Extended reality blends the physical and digital worlds, creating immersive experiences and computer-generated environments, in which users can interact with and manipulate digital objects or simulations. External portable devices like glasses and headsets often provide support, allowing to track movements and interact with virtual objects.

The **Artificial Intelligence (AI)** digital area encompasses the development, deployment, and uptake of AI techniques including but not limited to machine learning, deep learning, computer vision, natural language processing, and neural networks.

The **5G & Connectivity** digital area encompasses the facilitation of devices, machines, and systems to communicate and share data seamlessly, supporting a fast and reliable means of connectivity. Enhanced connectivity is ensured by the latest generations of wireless network technology (i.e. 5G and 6G), which offer faster data transfer speeds, lower latency, and the capacity to connect a massive number of devices simultaneously, essential to the integration of advanced technologies. The 5G & Connectivity digital area enables the implementation of Internet of Things (IoT) by fostering the connection and communication across devices and sensors, as well as gathering and storing generated data.

3D printing, or additive manufacturing, is a manufacturing process that uses special printers to build three-dimensional physical objects from digital designs. 3D Printing is a dual-use technology that finds applications in various manufacturing sectors, such as aerospace, automotive, and consumer goods.

Blockchain and distributed ledger technologies (DLT) represent a broader category – including blockchain – of decentralised systems where data is stored and synchronised across multiple nodes or computers in a network, without relying on a single authority for data management. They represent secure, transparent, and decentralised methods for recording, storing, sharing and managing data and transactions.

The **Internet of Things (IoT)** is a network of interconnected physical devices, machines, and objects that are embedded with sensors, software, and connectivity capabilities. Often complemented by Automatic Identification and Data Capture (AIDC) technologies (i.e. barcodes, RFID, QR codes), these devices are able to ‘sense’ their environment, identify, collect, store data from objects and products, and exchange them over the internet or other networks via standard communication protocols and without human intervention (i.e. machine-to-machine communication).

Online Platforms, Internet & Cloud:

Online platforms include websites, web applications, marketplaces, social media, and mobile apps. Internet & Cloud comprises of digital environments that enable interactions, transactions, and exchanges between individuals, business, or organisations through the Internet. This digital area also includes IT solutions that leverage cloud-computing infrastructures to give ubiquitous and on-demand network access to scalable computing resources, such as storage capacity and processing power for large volumes of data, software applications, and a range of services accessible over the internet - hosted on remote servers (data centres) by cloud service providers.

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